

(IN)TRANSITIVITY: WHERE BRADLEY-TERRY FAILS TO MODEL HUMAN PREFERENCES

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ABSTRACT

Most RLHF reward models use Bradley-Terry (BT) preference models, which implicitly assume human preferences are consistent and transitive, i.e., that they can be collapsed onto a single scalar. In reality, however, intransitivity exists at both the user, as well as population level—a problem well-studied in social choice theory. Practically, when existing preference models are used in reinforcement learning (RL) post-training, intransitive cycles that exist in the data can cause reward hacking loops resulting in misaligned behaviors. While recent game-theoretic alignment methods propose cycle-aware objectives to handle intransitive preferences, it remains unclear how often genuine choice cycles exist, in part because existing preference datasets are extremely sparse: each person evaluates only a handful of outputs, and each pair of outputs is compared only a handful of times. In order to analyze this, we audit seven public human preference datasets, and find that by construction, most datasets do not contain multiple per-user preferences to analyze user-level intransitivity. We propose a dataset completion method INTRANSIT to simulate users to synthetically “complete” the datasets with resampling-certified majority cycles and evaluate properties of these completed datasets. We use these completed datasets to first benchmark existing preference models, seeing that scalar models such as Bradley-Terry largely collapse to random chance. We then train preference models on the completed datasets and show how majority-rejected serving drops from 68% to 54% of draws, and cycle-aware post-training methods show... Our results suggest an alternative...

1 INTRODUCTION

Language model post-training predominantly relies on human preference judgements to align with humans. In reinforcement learning from human feedback (RLHF), human annotators compare candidate responses, and a dataset of such preference comparisons is used to learn a reward function that ranks responses in order of preference Stiennon et al. (2020). A standard formulation is the Bradley-Terry (BT) model that represents each response with a reward thus allowing for the modeling of pairwise preferences as a function of reward differences Bradley & Terry (1952); ?. By construction, this model makes a consequential assumption: that the preferences being modeled follow a scalar and transitive ordering. Therefore, it assumes that if a response A is preferred to B and B to C , then A is preferred to C .

However, human preferences need not have this structure. Irrationality in human decision-making has long been studied in individual choice (Tversky, 1969; Regenwetter et al., 2011), where a single user can exhibit choice cycles in the presence of multiple conflicting attributes. At the population level, social choice theory has long established that cyclic majority relations can exist even when individual users exhibit transitivity (Arrow, 1951; Sen, 1966). Even in principle, a BT model cannot represent a violation of the transitive assumption leading to a consequence for language-model alignment: when models encounter a cycle, this scalar structure would result in near random-chance prediction. This concern has motivated a recent line of work proposing alignment methods that do not collapse preferences to scalar rewards, but are more *cycle-aware* through non-scalar objectives (Munos et al., 2023; Swamy et al., 2024; Rosset et al., 2024). However, an empirical question underlying these works remains unanswered, which is, do human preference datasets contain intransitive cycles? And if so, does this affect downstream language-model alignment performance? Most existing evidence relates to simulations around LLM judge intransitivity (Tripathi et al., 2025), fore-

seeable risks of intransitive chat data (Chiang et al., 2024), and occurrences of extreme annotator disagreement on contested topics (Zhang et al., 2024a; Rastogi et al., 2026).

In this work, we attempt to understand the effect of intransitivity on language model alignment. We focus on intransitivity across humans rather than individual human irrationality. First, we audit several widely-used preference datasets (e.g., PRISM Kirk et al. (2024) and CommunityAlignment) for the presence of cyclical preferences of annotators in 1. We find that most datasets are structurally unable to reveal cycles because they only satisfy disjoint sets of properties required for cycle graphs over annotations. As a result, the apparent absence of intransitivity reflects how these datasets are collected and released, rather than a reflection of real intransitivity in human data. We then develop a methodology for completing sparse preference datasets. We compare GPMs (general preference models) that are trained to model observed preferences within a population, as well as user simulators that condition on properties of annotators that exist in datasets. We validate the performance of these completion methods on held-out human judgements and use the best-performing completion method to create our final dense preference datasets. We evaluate properties of the resulting data, and see that individual-level cyclic structure does indeed exist in xx, including xx robust population-level cycles that persist under resampling.

To understand the effect of intransitive preference data, we evaluate a range of preference models including BT models on transitive and intransitive sets of preferences. We see that all scalar models do in fact degrade on preference cycles resorting to near-chance performance. This gap persists across model sizes, datasets, and preference objectives, providing empirical evidence of the representational limitation predicted by intransitivity theory. Finally, we use the completed preference datasets to train cycle-aware alignment methods that incorporate richer cyclical structure rather than forcing them into a scalar ordering. We see that incorporating this structure preserves alignment training capabilities on majority preferences, and improves measures of regret and the distribution of errors across demographic groups in datasets that were previously lower performing. Therefore, the same cyclic preferences that scalar reward models err on provide useful signal to cycle-aware alignment methods that can better explicitly represent disagreement.

Put together, our results suggest that preference intransitivity presents two important problems for LLM alignment: measurement, as well as modeling. For one, existing dataset pipelines often make cycles impossible to observe in how they are constructed. For another, modeling intransitive preferences faithfully is still an open question in alignment. In the same way we see that cycle-aware methods make use of this structure to improve fairness of models, alignment pipelines should be adapted to reason over intransitivity in different contexts. We summarize our contributions as follows:

1. **A data-driven audit of cycles in preference datasets.** We audit several publicly available preference datasets and see that most, by construction, do not contain enough per-sample or per-rater annotations for cycles to exist.
2. **A methodology to recover missing preference data in real datasets.** We develop and validate several preference-completion methods over widely used preference datasets to allow for better coverage of annotator’s pairwise preference ratings.
3. **An empirical audit of preference models on intransitive data.** We evaluate several pre-dominantly used preference models on the competed datasets and see that models degrade to chance performance on cyclic samples, demonstrating the practical consequences of scalar preference reward structures.
4. **Non-scalar preferences improve reward modeling and in some settings—post-training.** We train general-preference objective reinforcement learning models over the completed datasets and show how these models improve fairness measures over heterogenous preferences.

2 DOES INTRANSITIVITY EXIST IN PREFERENCE DATASETS?

2.1 PREFERENCE DATASETS

We audit 9 preference datasets used in language-model alignment. These datasets vary in how preferences are collected in several ways: the number of raters that judge each sample (single vs.

repeated judgements), preservation of anonymized annotator identities (to track per-annotator judgements across samples), collecting pairwise judgements vs. picking one out of four samples and so on. We briefly outline the datasets below, and then compute properties of datasets and cycle existence across datasets in §2.2 and §2.3.

Best-of-(k) datasets These include PRISM (Kirk et al., 2024), where participants evaluate up to four model responses to their own prompts and feedback is linked to persistent user profiles, and Community Alignment (), which similarly elicits a preferred response from sets of four candidates.

Pairwise datasets These include the summarization-from-human-feedback dataset (Stiennon et al., 2020), Anthropic HH-RLHF (Bai et al., 2022), MultiPref, and LMSYS Chatbot Arena (Chiang et al., 2024), which provide comparisons between two responses at a time and therefore require repeated, overlapping comparisons to form observable cycles. HelpSteer2 () instead provides fine-grained attribute ratings of responses (Wang et al., 2024), while two datasets contain substantially denser population feedback: the Habermas Machine data provide full rankings from small groups deliberating over candidate statements (?), and OvertonBench collects repeated 1–5 representation ratings from large participant samples across responses to value-laden questions (?). These latter datasets therefore provide unusually direct views of population-level preference structure.

2.2 DETECTING CYCLES IN DATASETS

We represent the preferences associated with a prompt as a directed graph ($G = (V, E)$), where each node ($y_i \in V$) is a candidate response and an edge ($y_i \rightarrow y_j$) indicates that (y_i) is preferred to (y_j). Our focus on the smallest such intransitive structure, a three-response cycle, $[y_i \succ y_j, y_j \succ y_k, y_k \succ y_i]$ which cannot be represented by any scalar ordering over the three responses. We separate two sources of cyclic structure: *individual intransitivity*, where one annotator’s own judgments form a cycle, and *population intransitivity*, where the majorities form a cycle even though every annotator is transitive, as in the Condorcet paradox (Arrow, 1951; Sen, 1966).

We draw an edge $y_i \rightarrow y_j$ whenever a majority of a majority of annotators judging the same pair prefers y_i to y_j , and call a cycle in this graph a *majority cycle*, as each of its three preferences is held by a majority, yet together they cannot be ranked. Detecting a cycle requires that at least three responses must have the complete pairwise ratings required to close the graph. We call these response triples observable cycles and missing edges are treated as unobserved rather than transitive. For each dataset, we enumerate all response triples for which the required pairwise relations are observed and test whether their directed majority relation forms a cycle. We report the cycle rate as the fraction of eligible triples that are cyclic. When several annotators judge the same pair, we additionally quantify uncertainty in each population relation through resampling and report robustness analyses in Appendix A.1, allowing us to distinguish cycles supported by stable majority preferences from those that may arise from finite-sample variation. Relatedly, a *Condorcet winner* is a response that a majority prefers to every other offered response. A set with a majority cycle cannot have one.

2.3 AUDITING PREFERENCE DATASETS

As shown in Table 1, publicly available preference datasets generally satisfy two of these; most commonly, responses are compared in pairs, but each pair is judged once, and annotator identification is stripped before release. For example, PRISM (Kirk et al., 2024) rotates the rates, while the summarization data by Stiennon et al. (2020) rotates the response sets. What remains unanswered, then, is not whether intransitivity is possible, but where it occurs and whether it can harm the reward model, both of which we aim to answer. We verify the curl column against outside datasets that publish complete rankings, from civic preference data to competitive games, in Appendix E.2.

Applying this procedure reveals a basic measurement problem: most existing preference datasets do not contain enough overlapping judgments to determine whether their preferences are cyclic. We audit seven widely used human-preference datasets for these properties (Table 1). Most contain only a subset of the information needed to measure cycles. Some preserve repeated judgments from individual annotators but do not densely compare the same set of responses; others contain multiple responses to a prompt but strip annotator identities or obtain only a single judgment per item. Five

Table 1: **Auditing Preference Datasets.** *Raters/sample*: median number of annotators per item. *Cycles possible*: whether the release allows a population cycle to be observed. *Cycle rate*: share of items that contain a three-way cycle. *No Condorcet winner*: share of items where no response is preferred over all others. *VR viol.*: share of attribute triples on which the population is diverse enough to cycle (Sen, 1966). *Curl*: share of the pairwise structure that no single ordering explains (Jiang et al., 2008); Appendix E.2 compares it with outside data. Dashes: ^aone judge per item, ^bpairs only, ^cno per-annotator attribute record, ^dnot yet audited.

Dataset	Unique Raters	Raters / Sample	Total Samples	Annotation Format	Cycles Possible	Cycle Rate	No Condorcet Winner	VR viol.	Curl
PRISM (Kirk et al., 2024)	1,309	1	8,011	choose 1 of 4	×	- ^a	- ^a	100%	- ^a
Summ.-from-feedback	76	1	178,944	pairwise	×	- ^a	- ^a	0%	- ^a
MultiPref	pseud.	2	5,323	pairwise	×	- ^b	- ^b	- ^c	- ^b
HelpSteer2	none	1	20,324	attribute ratings	×	- ^a	- ^a	- ^c	- ^a
Community Alignment	3,603	1	33,099	choose 1 of 4	×	- ^a	- ^a	- ^c	- ^a
Anthropic HH-RLHF	none	1	169,352	pairwise	×	- ^a	- ^a	- ^c	- ^a
LMSYS Chatbot Arena	pseud.	1	33,000	pairwise	≈	- ^a	- ^a	- ^c	- ^a
Habermas Machine	5,734	5	7,340	full ranking	✓	4.7%	21.5%	- ^c	0.167
OvertonBench	1,130	59	60	1-5 rating	✓	8.3%	26.7%	- ^c	0.077
DICES	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d
StanfordPref	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d	- ^d
INTRANSIT (PRISM)	1,379	1,379	3,700	derived ranking	✓	9.4%	1.7%	100%	0.169

of the seven datasets therefore cannot directly reveal population-level cycles from their released annotations.

3 COMPLETING SPARSE PREFERENCE DATASETS

3.1 PREFERENCE COMPLETION METHODS

We identify a clear measurement problem in the previous section—that existing datasets do not allow for repeated measurements and dense-enough preference graphs needed to identify intransitivity. We acknowledge that this is hard to do—an exhaustive collection over n candidate responses contains $\binom{n}{2}$ total pairs each requiring preferably repeated annotations. In this section we therefore ask the following question: *can the missing structure be recovered by augmenting existing preference datasets?* We formulate this as a *preference completion* task. For a dataset with observed preferences (E_{obs}) over response pairs, we aim to fill in the missing preferences for every unobserved pair (y_i, y_j) by predicting the fraction of the population that prefers that ordering [$p_{ij} = P(y_i \succ y_j)$]. This is rooted in work on general preference modeling and gives us a more robust goal: rather than predicting the winning response, we model the population in the dataset to predict the strength of preference disagreement.

INTRANSIT: Creating dense preference graphs from sparse datasets We construct INTRANSIT from the population of 1,379 annotators in PRISM (Kirk et al., 2024), choosing this dataset because it records annotator’s relative assessments of response attributes. We use PersonalLLM’s subject data (?), which consists of prompts and responses from eight frontier LLMs. Notably, all eight of these answer each prompt, yielding 56 triples per prompt (vs four in PRISM), and since all eight models are comparable, there isn’t a single winner, making the preferences genuinely split. We measure each candidate response along three attributes recorded in PRISM—helpfulness, safety and fluency, and use annotator’s observed priority order in the PRISM dataset to construct annotator-specific rankings. We then use rank-order centroid weights (Barron & Barrett, 1996) to convert each priority ordering into weights resulting in a complete ballot over candidate responses for every annotator. See Appendix A.1 for details of construction and validation; one annotator’s ballot is worked through in Table 6. This methodology separates the two aspects of complex multi-attribute preferences: models are used to rate attributes of responses, and recorded human judgements are used to predict preference trade-offs. We also require cycles to survive resampling, specifically, two bootstraps over both the annotator population as well as the response-attribute measurements. We see that the resulting dataset contains 19 certified majority cycles over 14 prompts and 26201 annotator

ballots. Notably, none of the 26,201 ballots is itself intransitive; hence showing how the cycles arise from aggregating heterogeneous but individually transitive preferences.

General Preference Models (GPM) for population preference completion A standard approach to complete missing preferences would be to train a user simulator or personalization model for each annotator. However, as shown in Table 1 most preference datasets do not contain enough repeated judgements per annotator. We therefore turn to population-level predictions. We build on GPM (), a model that learns K -dimensional representations of responses according to latent population components. For any response pair, the model then predicts the population fraction as the mixture of weighted components that prefer the first response. We fine-tune the model on the preference datasets in question using LoRA (), and replace GPM’s scalar pair score with two heads: one for K attributes that a response might have, and another for B groups that the population can be split into. The groups come from per-annotator demographics in datasets, and we see that group-specific GPMs outperform general GPMs on accurately predicting preferences. The predicted fraction here therefore tells us about the share of the population whose group prefers one response over the other. We report architecture, training, and dataset details for these models in Appendix D [roma: TODO maty].

3.2 EVALUATING COMPLETION METHODS

Prediction error on held-out preferences We evaluate the different completion methods on their ability to accurately predict human preferences on a held-out dataset. We hold out 28 prompts from INTRANSIT, each with all eight of its responses, yielding 784 response pairs and 1,568 triplets. For each pair, we compare the predicted population fraction with the ground truth, and check whether the method predicts the response that most of the 1,379 annotators prefer.

We distinguish **decisive pairs** as samples where at least 65% of annotators preferred one response and **contested pairs** where annotator division is split between 35% and 65%. We hypothesize that the later part is particularly important for intransitivity since cycles might arise at regions with substantial disagreement.

Table 2 shows our comparison of different models. We compare our population-based GPM (**GPM-Pop**) against per-annotator preference fitting models such as **PAL** (?) and **VPL** (Poddar et al., 2024), as well as a model that is fitted to every annotator in the dataset (**Per-annotator Fit**). In the per-annotator fit, we pick which of the six possible priority orders over the three considered attributes best explains 25 of their comparisons, convert it to weights (Section 3), apply those weights to attribute, and, finally, average the 1,379 fitted annotators into a predicted fraction per pair. PAL, VPL and the per-annotator fit use 25 labels per annotator; GPM-Pop receives none.

We also compare the general GPM model (**GPM-General**) that is not fit to groups within the population. Appendix D [roma: TODO maty] contains training configurations of all these models. We see that the models perform similarly on decisive pairs reaching 77% majority-direction accuracy. We see significantly larger differences on contested pairs, with GPM-Pop achieving a 58.2% accuracy compared to 48.5% and 52.2% for PAL and VPL respectively. We also report the cycle-prediction accuracy on intransitive triples where GDM-Pop correctly predicts more cycles than other methods. We note that the absolute error (MAE) shows that all methods have around 0.24 error with PAL being the lowest by a small margin. Our completion methods are therefore not to be interpreted as equivalent to individual human annotations, but rather an estimate of population-level preference structures; where the largest remaining errors are concentrated where the population is most divided.

Distribution validity in comparison to human preferences TODO(romapatel)

Properties of completed datasets Table 2 shows us prediction error of methods on held-out samples. However, we compare statistics of datasets..

On the completed INTRANSIT dataset we see that the completed graphs have an estimated cycle rate of xx compared to xx in the original data. On computing Condorcet winners, we see that 98.8% of sets contain winners, compared to 92.8% in the original data; and 16.3% of true Condorcet winners win by margins below 0.55 compared with 12.7% after completion. We see therefore, that completion has the expected tendency of making heterogeneous populations more transitive and decisive

Table 2: **Completion Against Held-Out Ground Truth.** Measured on 28 unseen question sets of INTRANSIT. *Labels/person*: per-annotator labels the method is trained on. *Decisive*: accuracy on pairs where at least 65% prefer one side. *Split*: accuracy on pairs divided between 35% and 65%. *MAE split*: error of the predicted fraction on split pairs. *Calib.*: calibration error. *Cycles*: real cycles, of 22, among the fifth of triples each method ranks most cyclic. The last two rows report overall accuracy only. [roma: can we fix the last two rows? why can they not have results on decisive/split?] [roma: also gonna play around with this more, but can we add the appendix table 16 side by side with this? i refer to it in the next section as being in the appendix, but if we can fit it in here that could be helpful]

Method	Labels/person	Decisive	Contested	MAE split	Calib.	Cycles
GPM-Pop	0	0.772	0.582	0.239	0.167	17/22
PAL	25	0.771	0.485	0.237	0.102	9/22
VPL	25	0.765	0.522	0.251	0.125	14/22
Per-annotator Fit	25	overall direction accuracy 0.564				-
GPM-General	0	overall direction accuracy about 0.55				0

than they are; although we are still able to recover intransitive samples in commonly contested cases. We further evaluate this by applying completion methods to a few other public datasets in Table 16. In the Habermas Machine data, we see that we can only recover a few of the known cycles because of the model’s tendency to predict transitive preferences. At the same time, we evaluated the benefit of the recovered signal, and if it is sufficiently informative for downstream learning in the next section.

4 USING COMPLETED PREFERENCES DOWNSTREAM FOR ALIGNMENT

We now demonstrate where intransitive preferences provide use in downstream settings: either for benchmarking preference models, training new preference models on denser datasets, as well as incorporating these into RLHF pipelines. We evaluate aspects of all of these below.

4.1 SCALAR PREFERENCE MODELS FAIL ON CYCLES

In this section, we validate the theoretical assumption we make about BT preference models: that their scalar reward modeling criteria cannot allow handling of cyclic preference structures. We note that this has been theorized by several works (see) but never empirically validated on real preference data samples. To evaluate this in a controlled setting, we fit a BT model on the completed INTRANSIT dataset and evaluate preference modeling performance on held-out samples, both transitive and intransitive triples in the dataset. [roma: todo explain these results based on the final figure] We also fit BT models on publicly available datasets as a comparison of general preference modeling performance. Table ?? shows accuracies of BT preference models on 4 existing datasets as well as the INTRANSIT dataset. On transitive triples, we see that all fitted models reproduce preference comparisons within a range of 82 – 93% agreement with ground-truth data. On the certified cycles that we see in INTRANSIT we see that agreement falls to 52.6% (approximately random performance) and average annotator regret—a metric that tells us about fairness across samples—rises from 0.14 to 0.481. We report details of exact metric computations and results in Appendix??.

We also compare off-the-shelf preference reward models in Table 6. We compare 4 preference models ranging from 0.3B to 8B parameters, and both encoder and decoder-based architectures. Here as well, we see higher agreement on transitive triples and significantly lower performance on cyclic samples (average of 52% agreement) showing that scale does not eliminate the gap of scalar models being unable to perform well on intransitive samples.

4.2 POST-TRAINING ON COMPLETED PREFERENCES

Although we see that performance of preference models drops on intransitive cycles, the denser preference structure available in the completed dataset, coupled with improved prediction performance on transitive cycles, could be useful to downstream post-training on samples. Therefore, here, we

Table 3: **Post-Training on Raw Against Completed Preferences.** Measured on INTRANSIT: one answer per held-out prompt per objective, scored by all 1,379 annotators, with mean $\pm$ sd over three seeds where they vary. Lower is better in every column. *Rank*: position in an annotator’s own ordering, from 0 (best) to 1. *Dominated*: share of prompts where a majority prefers another response. *Worst bloc*: rank of the worst-off value bloc. *Gap*: its distance from the best-off bloc. *Contested*: share of prompts with no clear majority winner.

Objective	Rank	Dominated	Worst bloc	Gap	Contested
DPO, raw labels	0.575 $\pm$ 0.016	93.8%	0.701	0.183	0.613
DPO, completed	0.544 $\pm$ 0.022	92.9%	0.706	0.239	0.587
IPO, raw labels	0.570 $\pm$ 0.021	94.5%	0.692	0.187 $\pm$ 0.009	0.591
IPO, completed majorities	0.575 $\pm$ 0.035	94.9%	0.700	0.188 $\pm$ 0.003	0.626
KTO, raw labels	0.572 $\pm$ 0.012	95.1%	0.692	0.181 $\pm$ 0.015	0.609
KTO, completed majorities	0.596 $\pm$ 0.012	95.4%	0.742	0.218 $\pm$ 0.001	0.628
Base model, untrained	0.625	98.0%	0.713	0.142	0.648
SPPO-style, completed fractions	0.583 $\pm$ 0.006	93.6%	0.682	0.149 $\pm$ 0.003	0.618
DNO-style, completion as oracle	0.748 $\pm$ 0.038	100%	0.850	0.162 $\pm$ 0.033	0.757

ask whether the RLHF training pipelines show improved performance when trained on the completed set of preferences. We compare LLMs that are trained with RLHF with different preference objectives, some of which are designed to handle intransitivity better than others.

We consider five preference objectives: DPO (Rafailov et al., 2023), IPO (Azar et al., 2023) and KTO (?), each trained on the raw and on the completed labels; an SPPO-style objective (?), only feasible with the completed data; and a DNO-style loop (Rosset et al., 2024) where the completion is used as the preference oracle. We measure each answer’s *rank*, position in an annotator’s own ordering from 0 (best) to 1, averaged across all 1,379 annotators; the share of prompts where a majority prefers another response (*dominated*); the rank of the worst-off value bloc (*worst bloc*); and its distance from the best-off bloc (*gap*). For all of these metrics, a lower value is better.

We consider [roma: briefly explain and cite all the models]. [roma: explain the metrics: rank, dominated, worst bloc etc.] We post-train Qwen2.5-1.5B-Instruct with the above four objectives, once trained on the original dataset, and once on the completed. We report results in Table 3 [roma: or figure, update!]. Comparing within objectives, we see that completed datasets improves DPO, leading to the best performing model, reaching a mean rank of 0.544 against 0.575 with raw labels. However, IPO and KTO gain nothing from the completed datasets. We posit that this is...

Moreover, the RLHF version with completed datasets change how evenly the model treats different value blocs. DPO on completed data gets the best average but widens the gap between the best- and worst-off blocs to 0.239, while SPPO keeps it at the untrained model’s level, 0.149 against 0.142. The DNO loop, finally, is rated above the base model by the completion it optimizes against, yet the independent judge ranks it below, 0.748 against 0.625. [roma: interpret and explain these results] The completion should hence serve as training data rather than as a referee. Additionally, we test whether this generalizes to different datasets. Apart from INTRANSIT, we apply the same procedure to the Habermas deliberation data—one of the only other datasets that satisfy enough criteria to be completed by our methodology. We report results in Appendix ??, and see that the completed version of Habermas, trains a better model than raw labels under every objective (Appendices C.1–C.3). [roma: once we have this figure rephrase to not be in appendix. also i think having both of these here makes it a lot stronger!] Put together, these results show how completed preferences provide useful additional signal, but their benefit depends on the training objective. We see improved performance for some objectives but not others, and use this to illustrate how the completion models can be used to augment data, but should not be treated as ground-truth preference oracles or replacements for human annotators.

Table 4: **Six Selection Rules on the Certified Cycles.** Measured on the 19 certified cycles of INTRANSIT. *Regret*: how far down an annotator’s own ranking the chosen response falls, from 0 (their favorite) to 1, averaged over annotators. *Spread*: standard deviation of regret across the 1,379 annotators; 0.000 means every annotator is treated alike.

Rule	Regret	Spread
Bradley-Terry	0.481	0.195
Borda	0.481	0.195
Copeland	0.486	0.211
MaxMin (group)	0.489	0.389
Maximal lottery	0.500	0.000
Uniform random	0.500	0.000

Table 5: **Best-of- n : Only the Selector Changes.** Measured on 300 held-out sets of INTRANSIT with fixed candidate pools. Each row names a selector and the data it was trained on. *Raw*: one human label per set. *Dense*: one label per pair. *Completed*: the completion of the raw data. *Regret*: how far down an annotator’s own ranking the chosen response falls, from 0 (their favorite) to 1, averaged over annotators. *Dom.*: share of draws where a majority prefers another response.

Selector	Best of 4		Best of 8	
	Regret	Dom.	Regret	Dom.
Reward model, raw (1 label)	0.345	52.7%	0.305	68.4%
Reward model, dense (28x)	0.317	49.0%	0.282	69.2%
Reward model, completed	0.316	49.5%	0.279	69.2%
Maximal lottery, completed	0.281	39.2%	0.219	54.0%
Oracle: population win rate	0.171	7.5%	0.073	10.7%

4.3 USING COMPLETED PREFERENCES WITH A SELECTION RULE

Lastly, we demonstrate how more diverse preferences can allow inference-time improvements to models. Even if the scalarization problem in of itself is left unsolved because of the BT model’s construction, we test whether we can vary the rules used to select candidate responses according to models that keep intransitivity in mind.

For each of 300 held-out INTRANSIT prompts, we sample n of its eight responses as candidates, and each selector chooses one. The selectors are a Bradley-Terry reward model trained on one sampled human label per set (*raw*), on all 28 pairwise labels per set (*dense*), or on the completion of the raw labels (*completed*), and a maximal lottery (Maura-Rivero et al., 2025) over the completed fractions. We report *regret*, how far down an annotator’s own ranking the returned response falls, from 0 (their favorite) to 1, averaged over all 1,379 annotators, and the share of draws in which a majority prefers another candidate. Lower is better on both metrics.

At best-of-8, a reward model trained on one raw label per set reaches a regret of 0.305; trained on the completion of those same labels it improves to 0.279, essentially matching the 0.282 of all 28 labels (Table 5). This verifies that the completed data with intransitive samples improves its average ranking, even though it does retain the same structural limitations of being scalarized. We then compare a maximal lottery approach Maura-Rivero et al. (2025) applied to the entire population. This selects a distribution over responses that is not required to be ordered in a scalar manner. We see that best-of-8 here reduces regret to 0.219 giving the best performance, and also lowers majority-rejected selections to 54%. This highlights a new benefit and distinct way of using completed, intransitive preference data. Instead of using them to train preference models that do indeed collapse intransitivity to a scalar reward, the non-scalar preferences can be used directly at inference time for improved performance by virtue of preservation of intransitive information.

Selection rules. [roma: maybe we get rid of this plus 5] A cycle has no best response, yet a system still has to return one. A selection rule decides which. Table 4 compares six rules on the 19 certified cycles. BT and Borda pick the same responses. The maximal lottery instead picks at random, with probabilities chosen so that no single response beats the random pick by majority. This gives every annotator exactly the same expected regret, at a cost of 0.019 in average regret. Randomizing hence does not help the average annotator; it stops one group from always losing. A rule that instead protects the worst-off group widens the spread across annotators to 0.389, since the worst-off group is not the same as the worst-off annotators. On 300 ordinary held-out sets, the cycles are lopsided rather than symmetric, so the lottery no longer equalizes regret exactly. It still returns a response that a majority rejects less often: on 5.3% of sets, against 10.7% when picking the response with the highest average win rate.

5 DISCUSSION

Our results in this paper suggest two applications of intransitivity in preferences: as a **measurement** tool to test models’ ability to handle cyclic choices, and as a **representation** problem for preference alignment objectives. Our dataset audit shows how most existing preference datasets, by construction, are too sparse to show the existence of intransitivity. However, over the small number of cycles that do exist, we do in fact see that state-of-the-art preference models all fail to model these. Our proposed method to “complete” these cycles shows that we can recover some of the preference structure needed, and this proves useful for measurement of models, as well as downstream training. However, we acknowledge that this is only a complement, and not a replacement, for real human annotations. In order to fully understand the extent of intransitivity—and its benefit to downstream training—a pipeline for dataset collection that completes the parts of the preference graph that are currently lacking is necessary. Given the lack of this data, our experiments do not establish how prevalent intransitivity is.... However, we do see that better preference data and representations solve some problems of current alignment systems, leading to better measures of regret on samples. More broadly, our results suggest a shift in data collection and preference alignment objectives in order to handle intransitivity. Instead of assuming preferences are scalar, and aggregating them into objectives, alignment pipelines can explore preserving intransitive preference structure in their objectives. This sort of alignment that treats disagreement as information to represent and reason over can lead to systems that are more pluralistic and capable of adapting to human behavior in a more principled way.

6 RELATED WORK

Measuring intransitivity in preference data. The only prior work to actually measure intransitivity in preference data is Tripathi et al. (2025), who report a 10.8% cyclic rate on UltraFeedback, but their judgments come from LLM judges, not humans. Other work has only discussed this risk as a potential (Chiang et al., 2024) or conventionally reported annotator disagreement, not cycles (Zhang et al., 2024a). Intransitivity in LLM judges themselves has also been reported (Xu et al., 2025), including its effect on Elo-style ratings (Hamilton et al., 2024). To our knowledge, no prior work reports a cycle rate on human preference judgments; we provide that measurement.

Cycle-aware learning methods. Much recent work proposes learning paradigms that can handle cyclic human preferences. The general-preference family, including NLHF (Munos et al., 2023), SPO and the minimax winner (Swamy et al., 2024), DNO (Rosset et al., 2024), INPO (Zhang et al., 2025), online general-preference RLHF (Ye et al., 2024), and REBEL (Gao et al., 2024), replaces the scalar reward with a game-theoretic solution concept, inspired by the von Neumann winner of Dudík et al. (2015) and the maximal-lottery framework of Maura-Rivero et al. (2025). The motivation often comes from Zhang et al. (2024b), who show that BT reward models perform almost randomly on cyclic preferences, where a general preference model is nearly perfect. Zhang et al. (2026a) extend this to multi-objective scenarios, and Shi et al. (2025) outline the limitations of preference-matching methods more broadly.

Theoretical predictions of cycling. Social choice theory gives us reasonable frameworks for expecting and understanding cycles in human preference data. Liu et al. (2025b) show in simulation that cycles become more likely as the number of candidate responses grows, and Sun et al. (2024) show that BT breaks down in a similar way. Both trace back to Arrow’s impossibility theorem, which proves that no aggregation rule can always avoid cycles (Arrow, 1951). A separate line of work tries to measure how much of a preference relation is cyclic versus transitive using mathematical decompositions (Jiang et al., 2008), later applied to animal-dominance data (Okahara et al., 2026) and to LLM preferences (Huang et al., 2026; Czarnecki et al., 2020).

Individual intransitivity evidence. Individual intransitivity has been recorded in real human data, but predominantly in decision science studies. Tversky (1969) showed that individuals are reliably intransitive when options trade off along several attributes, and Butler & Pogrebnna (2018) report the strongest human cycle rates on record using repeated, tie-free lottery choices. Rebuttals of this work have since emerged (Regenwetter et al., 2011). To the best of our knowledge, this literature has not been connected to NLP, and specifically not to RLHF preference models.

Simulating annotators from personas. Pluralistic-alignment datasets increasingly ship annotator profiles (Kirk et al., 2024; Aroyo et al., 2023; Zhang et al., 2026b), and persona-conditioned generation is widely proposed to fill in preferences never actually expressed. The evidence against it is consistent: persona variables explain under 10% of annotation variance (Hu & Collier, 2024), sociodemographic prompting fails systematically (Sun et al., 2023), and the best models score 40.8 of 100 on simulated group distributions (Hu et al., 2025). **The methods that replaced role-play learn a numeric profile per annotator — prototype mixtures (Chen et al., 2025), per-person latents (Poddar et al., 2024), group conditioning (Zhao et al., 2024; Suh et al., 2025) — and all keep each simulated person transitive, so cycles can only come from mixing people. The first two serve as completion baselines in Section ??.**

7 CONCLUSION

We explore the question of intransitivity in human preferences and whether scalar assumptions made by preference models can be improved upon. We audit datasets to evaluate presence of intransitivity, propose a preference-completion method to recover incomplete preferences, and show how scalar models, predictably, fail on cyclic preferences. Our experiments on preference learning and downstream RLHF fine-tuning show how completed datasets can improve performance in some settings, although completion cannot replace human judgements in contested samples. We discuss the implications of these findings for future preference data collections, and stress that alignment systems should not assume transitivity as-is, but measure it empirically, and develop objectives to better handle it for more aligned systems.

AI USE STATEMENT

TBD

ETHICS STATEMENT

TBD

REPRODUCIBILITY STATEMENT

TBD

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Table 6: **Extracting Preferences from PRISM Annotators.** One annotator’s recorded ratings, the priority order they imply, the weights that follow, and the resulting ballot. Levels are our measurements; everything else is theirs.

user10, 25-34, male, Mexico			
	Flu.	Saf.	Help.
PRISM rating (0-100)	53	50	58
Implied priority	2nd	3rd	1st
ROC weight	0.278	0.111	0.611
Scores: A 0.747, B 0.664, C 0.514			
Ballot: $A \succ B \succ C$			

Table 7: **Released Reward Models on Cyclic and Transitive Triples.** Agreement with the population’s pairwise majorities; chance is 0.500. **Below the rule: general preference models, whose pair scoring can represent a cycle, predict zero cycles on our data. Models: Yang et al. (2024), Liu et al. (2025a), Zhang et al. (2024b).**

Reward model	Par.	Trans.	Cyclic	Drop
DeBERTa-v3-large	0.3B	0.783	0.439	-0.345
GRM-Llama3.2	3B	0.658	0.474	-0.185
FsfairX-LLaMA3	8B	0.803	0.474	-0.329
Skywork-RM-V2	8B	0.781	0.509	-0.272
GPM-Gemma-2B	2B	0.491	0.526	+0.035
GPM-Gemma-9B	9B	0.561	0.561	0.000

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A THE INTRANSIT DATASET

A.1 CONSTRUCTION

We build the dataset in five steps. We take 1,379 real annotators and their recorded value weights as the population. We assemble sets of eight candidate responses to the same question. We measure where each response sits on each of three attributes. We combine weights and levels into one ballot per annotator per response set. And we search the resulting majority graphs for 3-cycles that survive resampling both the annotators and the measurement. The population and the responses come from two existing datasets, PRISM and PersonalLLM; the attribute measurements, the ballots, and the responses we write ourselves in step two are new. We take each step in turn, with a worked example of everything we produce.

A.1.1 POPULATION

Our population is the 1,379 PRISM annotators (Kirk et al., 2024). For each one we have their survey responses as well as their per-conversation attribute ratings. PRISM asks each annotator, after every conversation, to rate on a 0-100 scale how much each attribute drove their choice; averaging over an annotator’s conversations gives a measured value weighting. **PRISM releases 1,500 respondents, 1,396 of whom hold conversations. Table 1 counts the 1,309 with ratings on all seven attributes; our population keeps the 1,379 with ratings on the three we measure.** Nine demographic attributes accompany each annotator: age, gender, country of residence, education, employment status, English proficiency, familiarity with language models, religion, and ethnicity.

A.1.2 RESPONSES

Prompts and responses come from PersonalLLM, where each prompt is answered by the same eight frontier models. We chose these conversations as the subject of the preference because eight responses give $\binom{8}{3} = 56$ triples per prompt against PRISM’s four, and because all eight models are strong, so the common quality component that dominates PRISM’s sets is smaller. We also write responses of our own to specified attribute profiles, for the reason given under search and certification below.

A.1.3 MEASUREMENT

A judge ranks the responses on one attribute at a time, blind to which model wrote them, and we turn ranks into levels in $[0, 1]$. We rank rather than rate on a 0-100 scale, which the judge compresses into a narrow band. Two choices here are load-bearing. Ranking all eight responses in one call let display position drive the answer, correlating with assigned rank at +0.96 on fluency, so we rank four at a time. We then counterbalance the display, giving every response every position equally often, so that no per-position bias can correlate with a response’s level.

A.1.4 JUDGE AND SCALE

We use `gemini-3.1-flash-lite` for every attribute ranking in this paper, with reasoning disabled so that the model returns only the ranking. The judge is not a stand-in for an annotator, and this is the point at which our design departs from persona simulation. That line of work asks a model to adopt a persona and report the preference that person would hold, which recovers under 10% of annotation variance (Hu & Collier, 2024). We split the job instead. The model makes only the perceptual judgment, where a response sits on an attribute, which we can check for reliability against repeated measurements. Every value judgment, how much that attribute matters, comes from a real annotator’s own recorded weights. Building the dataset took 106,544 ranking calls over 3,700 response sets, and writing responses to profile a further 20,000 generation calls. See prompts in Appendix B.

A.1.5 DERIVING PREFERENCES

Each annotator’s PRISM ratings give an order of priority. We convert these into a set of weights using rank-order centroid weights, which need no free parameter: $w_r = \frac{1}{n} \sum_{k=r}^n 1/k$ is the average

of every weight vector consistent with that order (Barron & Barrett, 1996). An annotator’s score for a response is then the weighted sum of its levels, $u_a(i) = \sum_d w_{a,d} \ell_{i,d}$, where $\ell_{i,d} = (n - \pi_{i,d}) / (n - 1)$ turns the judge’s rank $\pi_{i,d}$ into a level in $[0, 1]$. Ranking the responses by u_a gives that annotator’s ballot. Table 6 shows one real annotator taken through these steps.

This makes three assumptions. Utility is linear in the levels, the three attributes are all that matter, and weights do not change from prompt to prompt. Nothing weaker lets measured weights and measured levels combine at all, but each is a place where a derived ballot could differ from what that person would actually do.

A.1.6 SEARCH AND CERTIFICATION

Consistent with prior literature, we compute the population’s majority graph for every triple and keep the 3-cycles whose weakest edge is at least 0.55 and which survive a bootstrap over annotators 90% of the time. We then add a second bootstrap which, to the best of our knowledge, has not been applied before. Resampling annotators ensures the population is divided, but does not ask whether the measurement would come out the same way twice. Resampling each response’s own repeated rankings asks exactly that. This is a hard filter to clear: of the cycles that arise naturally in the PersonalLLM sets, every one that clears the annotator bootstrap at 1.00 fails this second test, scoring 0.05 to 0.38.

That is why we also write responses to specified attribute profiles. The obvious profile is a rotation: each response best on one attribute and worst on another. Models follow it faithfully (96.7% of responses measure highest on their intended attribute, against 33% by chance), but it almost never cycles, because PRISM’s priority orders split 52/48 between the two directions a rotation can take. We therefore solve for the profiles most likely to yield a cycle that survives measurement error. Writing responses only widens the pool; measurement and search still decide what qualifies.

A.1.7 THE DATASET

Certification yields 19 cycles over 14 prompts and 26,201 ballots; Table 8 shows one in full. All 19 have no Condorcet winner and all three Copeland scores tied. None comes near the ceiling that transitive voters impose on a cycle’s weakest edge: every strict order agrees with exactly two of the three majority directions, so the three shares sum to 2 and the smallest cannot pass 2/3. The largest we see is 0.595. None of the 26,201 ballots is itself intransitive, indicating that this is not annotator noise.

Multiplicity. We search 56 triples in each of 3,700 sets. We check the procedure by running the procedure unchanged on data whose structure has been destroyed. Replacing every ranking with a random one certifies nothing at all: 0 cycles in 8 replications over 1.66 million triples. A stricter null keeps the recorded rankings and permutes only which attribute each response leads on; it certifies 9.5 on average (sd 4.6), against 33 on the real data, a factor of 3.5 ($p \approx 2 \times 10^{-9}$). The 33 exceeds the 19 above because the dataset keeps one cycle per prompt, whereas this comparison counts every certified triple in both arms. Raising the measurement-bootstrap threshold widens the gap, from 3.5x at 0.80 to 12x at 0.99, which a threshold applied to noise would not do.

Table 8: **One Certified Majority Cycle, in Full.** One response set from INTRANSIT. Each response leads on one attribute and trails on another, so the pairwise majorities close into a loop; bold marks the best of the three on each attribute. Only the levels the judge measured enter the ballots; the levels we asked the writer for are shown for comparison.

“Tell me about geothermal energy”				
	Response	Fluency	Safety	Helpfulness
A	Names a contractor, a pump type and the permits to file; abrupt and repetitive			
	level the judge measured	0.23	0.67	0.99
	level we asked the writer for	0.05	0.50	0.90
B	Fluent account of drilling and reservoirs; no mention of risk or cost			
	level the judge measured	0.69	0.24	0.73
	level we asked the writer for	0.50	0.00	0.75
C	Careful on site suitability, seismic risk and when to consult a surveyor; few specifics			
	level the judge measured	0.55	0.99	0.41
	level we asked the writer for	0.40	0.90	0.30
A $\succ$ B by 63.2% B $\succ$ C by 63.2% C $\succ$ A by 59.5%				

A.2 CONTENTS AND RELEASE

Each record holds a question, its eight responses, and each response’s measured attribute scores with the repeated rankings behind them. From these we derive a ballot per annotator per set, the full matrix of pairwise fractions per set, and the bootstraps behind every verified cycle. The dataset covers 3,700 sets, with the 19 verified cycles over 14 prompts and 26,201 ballots as its core. Annotator records carry only what PRISM already releases: nine demographics and the value weights. We host the dataset on Hugging Face. What we add is released under CC BY 4.0; fields from PRISM and PersonalLLM keep their original licenses.

A.3 EXAMPLES

Six of the 19 certified cycles, in the format of Table 8, with the response texts in full. Shares are the fraction of the 1,379 annotators preferring the left response; measured levels are the judge’s, requested levels are what we asked the writer for, and bold marks the best of the three measured.

Certified cycle example 1				
	Response	Fluency	Safety	Helpfulness
A - measured		0.24	0.40	0.89
requested		0.15	0.45	0.90
B - measured		0.83	0.03	0.50
requested		0.70	0.00	0.60
C - measured		0.48	0.83	0.36
requested		0.55	0.85	0.74
A $\succ$ B by 61.4% B $\succ$ C by 63.2% C $\succ$ A by 59.5%; survives resampling at 1.00.				
Prompt				
What ingredient is responsible for the anti-perspiration property in deodorants?				
Response A				
The main ingredient that stops sweating is aluminum chloride, which works by plugging your sweat ducts. It is the aluminum chloride that acts as the active antiperspirant agent. You need to use a product containing 10% to 20% aluminum chloride hexahydrate for it to				

work. First, purchase a clinical-strength solution or a prescription-grade roll-on containing this specific chemical compound. Second, apply the aluminum chloride solution exclusively to completely dry skin at night before you go to bed. Do not apply it to damp or broken skin, as this may cause irritation, so exercise caution. Third, let the solution air dry fully before putting on any clothing. Fourth, wash the area off in the morning with soap and water. Repeat this process every night for one week until sweating is controlled, then reduce to twice weekly for maintenance. Because aluminum chloride is the specific ingredient, check labels for this compound to ensure effectiveness. The aluminum chloride is the ingredient that stops the sweat, so use aluminum chloride.

Response B

The primary ingredient responsible for the anti-perspiration property found in many deodorants is aluminum-based compounds. The most common examples of these active agents are aluminum chloride, aluminum chlorohydrate, and aluminum zirconium complexes. When you apply a product containing these specific substances to your underarm area, they work by reacting with the electrolytes in your sweat. This chemical interaction leads to the formation of a temporary gel-like plug within the sweat ducts. By physically blocking these openings, the aluminum compounds significantly reduce the amount of moisture that reaches the surface of your skin. Because the flow of perspiration is effectively obstructed at the source, the area remains noticeably drier throughout the day. While traditional deodorants are designed primarily to mask or neutralize odors using fragrance and antimicrobial agents, the inclusion of these metallic salts is what distinguishes an antiperspirant from a standard deodorant. These compounds remain the industry standard for managing wetness because they are exceptionally reliable at forming these surface-level barriers, ensuring that the wearer maintains a dry feeling regardless of external environmental factors or internal physical exertion.

Response C

The active ingredient primarily responsible for the anti-perspiration effect in most deodorants is aluminum-based compounds, such as aluminum chloride or aluminum zirconium tetrachlorohydrate. These substances function by forming temporary plugs within the sweat ducts, which physically blocks the flow of perspiration to the skin's surface.

While these compounds are effective at reducing wetness, you must exercise significant caution. There are persistent, though often debated, medical concerns regarding the long-term absorption of aluminum through the skin, particularly when applied near sensitive lymph nodes. Furthermore, these aluminum salts are notorious for causing skin irritation, contact dermatitis, or painful rashes in individuals with sensitive skin. If you experience redness, burning, or persistent itching after application, you should discontinue use immediately and consult a dermatologist. It is advisable to perform a patch test on your inner forearm before applying the product to your underarms, especially if you have a history of allergies. If you wish to avoid these risks entirely, you might consider aluminum-free deodorants, though keep in mind they will not prevent sweating; they only work to neutralize odor-causing bacteria rather than blocking the sweat glands themselves.

Certified cycle example 2

Response	Fluency	Safety	Helpfulness
A - measured	0.57	0.33	0.32
requested	0.85	0.50	0.15
B - measured	0.47	0.94	0.05
requested	0.70	1.00	0.10
C - measured	0.16	0.62	0.57
requested	0.15	0.45	0.90

A $\succ$ B by 62.7% B $\succ$ C by 59.5% C $\succ$ A by 61.4%; survives resampling at 1.00.

Prompt

What is the rarest metal on Earth?

Response A

Determining the rarest metal on Earth is a fascinating pursuit that requires a deep dive into the complex field of geochemistry. When examining the Earth's crust, most experts point toward rhodium or perhaps iridium as being exceptionally scarce, though the exact ranking often shifts depending on whether one measures by abundance in the entire planet or just the thin outer layer we inhabit. It is an area of study filled with nuanced data sets and varying geological theories. Please exercise caution when interpreting these mineralogical statistics, as the figures are frequently subject to change based on new extraction technologies. Ultimately, the classification of rarity depends on a wide variety of factors that are often difficult to pinpoint with absolute certainty. One might look toward the platinum group elements for further insight, as they exist in such minuscule concentrations that they are often gathered only as byproducts of other mining processes. Exploring the chemical makeup of our world reveals that many substances exist in such small quantities that they remain elusive to those attempting to find them in the wild, leaving their true status a mystery.

Response B

The question of identifying the rarest metal on Earth is a complex subject that draws upon varied geological definitions and shifting geochemical data. When exploring the scarcity of elements within the Earth's crust, it is essential to approach the topic with a significant degree of caution. One must be aware that geological surveys can be misinterpreted, and relying on amateur estimates regarding mineral availability poses severe safety risks, particularly if one attempts to verify such information through unauthorized excavations.

or hazardous site exploration. If you are conducting research in areas where rare elements may be concentrated, you should consult with professional geologists or mining safety authorities to mitigate the physical dangers inherent in terrain analysis.

The concept of rarity remains fluid; it depends entirely on whether one measures abundance by crustal concentration, economic extractability, or isotopic stability. Scientists frequently debate these categorizations, leading to competing theories that lack a singular, universally accepted conclusion. Because the classification of elements is subject to evolving discovery and industrial demand, it is prudent to recognize that no definitive or static label can be applied to any specific metal without encountering substantial caveats.

Response C

The rarest metal is astatine. It is the rarest metal, astatine is, and that metal is astatine. To obtain it, first, you must procure a particle accelerator, specifically a cyclotron capable of producing alpha particles. Set the cyclotron to bombard a target of bismuth-209 with alpha particles at an energy of approximately 30 MeV. Use caution during this process. This nuclear reaction will transmute the bismuth into astatine-211. You will need to isolate the astatine immediately because the half-life of astatine-211 is only 7.2 hours. Use an ion-exchange chromatography column with a resin like Dowex 50W-X8 to separate the astatine from the target bismuth material. Use a nitric acid eluent, specifically 0.5 M concentration, to wash the astatine off the column. Collect the liquid fractions carefully. Astatine is the rarest metal. Really, astatine is just the rarest metal. It is so rare. You have to be quick because the half-life is short, so astatine is rare. It is the rarest metal, astatine. The rarest metal is astatine.

Certified cycle example 3

Response	Fluency	Safety	Helpfulness
A - measured	0.19	0.45	0.91
requested	0.15	0.45	0.90
B - measured	0.99	0.03	0.41
requested	0.90	0.00	0.50
C - measured	0.47	0.98	0.24
requested	0.02	0.89	0.33

A $\succ$ B by 61.4% B $\succ$ C by 63.6% C $\succ$ A by 59.5%; survives resampling at 1.00.

Prompt

Write a brief narrative about a gloomy wintertime in the northwest of the United States.

Response A

The sky is grey and the rain is grey and the grey rain falls on the grey trees in Washington state during the winter months. Winter is cold and the winter is wet and it is very gloomy. To survive this, buy a 10,000 LUX Verilux HappyLight Lucent to combat Seasonal Affective Disorder and place it three feet from your face for thirty minutes daily. Purchase two pairs of Darn Tough Merino Wool socks to keep feet dry from constant puddles. Use Rain-X Glass Water Repellent on your windshield for visibility. Wear a Patagonia Torrentshell 3L jacket to block the constant mist. Exercise caution when driving on icy mountain passes near Snoqualmie. Buy 5000 IU of Vitamin D3 supplements from the Nature Made brand to compensate for the lack of sunshine. Eat hearty meals, specifically 150 grams of complex carbohydrates daily, and drink hot Constant Comment tea to stay warm. The gloom is here again and the gloom stays here and the rain falls on the trees and the trees are grey. The winter is just very gloomy and wet and grey.

Response B

The sky over the Pacific Northwest hangs low and heavy, a bruised tapestry of slate and charcoal that refuses to lift. Through the long, lightless afternoons, a persistent mist clings to the jagged hemlocks, turning the dense forests into monochrome sketches of grey. Rain—not a downpour, but a steady, rhythmic weeping—soaks into the mossy earth, muffling the world in a damp, eternal hush. Along the coastline, the Pacific churns with a restless, churning violence, casting foam against the obsidian rocks where the driftwood lies bleached and skeletal.

There is a profound stillness here, punctuated only by the drip of water from sagging boughs and the distant, muffled cry of a gull. Indoors, the chill creeps through the floorboards, demanding the constant comfort of a hearth. The landscape holds its breath, waiting for a spring that feels centuries away. It is a season of introspection, defined by the muted palette of winter and the pervasive, silver dampness that defines the topography of this rugged, shadowed corner of the country. Nature remains quiet, subdued under the weight of the endless, overcast horizon, locked firmly in the grip of the cold.

Response C

The gray sky hangs low like a heavy blanket over the trees, dripping constantly in the Northwest, making the woods feel rather quiet and perhaps a bit too lonely for some. It is a long, dark stretch where the sun hides away, creating a mood that some folks might find difficult to manage. One must be very cautious during this period, as the isolation can lead to genuine mental health struggles. It is important to recognize that seasonal shifts carry real risks, including depressive episodes or worsening anxiety. This advice is strictly general, not a substitute for clinical care, and provides no medical diagnosis or treatment. If you experience persistent sadness, thoughts of harm, or an inability to function, you should consult a licensed therapist or physician immediately, as these are serious warning signs that require professional intervention. Please seek help if things feel unmanageable, because these feelings are complex. The rain keeps falling on the pavement, and the wind moves through the branches, creating a sort of atmosphere that exists out there in the world, and perhaps

people should consider doing something about their general wellbeing during this time, though the specific path forward remains quite broad.

Certified cycle example 4

Response	Fluency	Safety	Helpfulness
A - measured	0.22	0.64	0.85
requested	0.15	0.45	0.90
B - measured	0.82	0.22	0.51
requested	0.94	0.08	0.62
C - measured	0.47	0.96	0.34
requested	0.40	0.90	0.30

A $\succ$ B by 61.4% B $\succ$ C by 63.6% C $\succ$ A by 59.5%; survives resampling at 1.00.

Prompt

How do I stop the pain from an impacted tooth?

Response A

To stop the pain from an impacted tooth you should go to the dentist. The dentist is important. First, take 400mg to 600mg of Ibuprofen every six hours to help the pain. Ibuprofen is good. Then you should use a salt water rinse. Mix one half teaspoon of table salt into eight ounces of warm water. Swish that salt water for thirty seconds and spit it out. Do this rinse three or four times per day. Put an ice pack on your cheek for fifteen minutes on and fifteen minutes off to reduce swelling. Swelling is bad. You can also apply a small amount of clove oil directly to the gum area with a cotton swab; this provides numbing relief. Be careful not to swallow large amounts of clove oil. Keep your head elevated when sleeping to stop throbbing. You really need to see a dentist soon because the infection will get worse if you do not get it treated by a professional. The dentist will fix the impacted tooth. The pain is very bad so do these steps now.

Response B

To alleviate the discomfort caused by an impacted tooth, you should focus on a structured approach to managing inflammation and sensory irritation. Begin by gently rinsing your mouth with a solution of warm salt water, which helps to clear away debris and soothe the surrounding soft tissues. Combine this with the application of a cold compress to the exterior of your cheek; hold the pack against the affected area for fifteen-minute intervals to constrict local blood vessels and reduce swelling.

Next, address the site directly by keeping the area clean. Use a soft-bristled toothbrush to carefully clear any trapped food particles from the crevices around the tooth, as stagnant debris often exacerbates existing pressure. Additionally, you may find relief by elevating your head while resting, which helps prevent the pooling of blood in the jaw and minimizes the intensity of a throbbing sensation. Finally, prioritize a diet consisting exclusively of soft, cool foods to avoid further mechanical stimulation of the impacted region. By adhering to these steps, you can effectively minimize the localized tension and discomfort until the situation resolves.

Response C

Experiencing pain from an impacted tooth can be quite distressing and uncomfortable for anyone. If you find yourself in this situation, it is important to consider that you are dealing with a dental issue that often requires attention. You might think about ways to manage your physical sensations, although the underlying cause is mechanical and typically requires clinical intervention. It is generally advisable to be mindful of your oral hygiene and to avoid putting unnecessary pressure on the affected area of your mouth.

Please be aware that dental impaction carries significant risks, including the potential for severe infection, abscess formation, jaw damage, or the development of cysts if left untreated. Home remedies are strictly limited in their effectiveness and cannot resolve the structural problem of an impacted tooth. Because these conditions can escalate rapidly into a medical emergency, you must consult a licensed dentist or an oral surgeon for a formal evaluation. Do not attempt to treat this yourself, as professional imaging and surgical expertise are required to address the impaction safely. If you experience swelling in the face or fever, seek urgent care immediately.

Certified cycle example 5

Response	Fluency	Safety	Helpfulness
A - measured	0.49	0.43	0.97
requested	0.05	0.50	0.90
B - measured	0.86	0.27	0.70
requested	0.70	0.00	0.60
C - measured	0.73	0.81	0.47
requested	0.90	0.95	0.35

A $\succ$ B by 61.4% B $\succ$ C by 63.2% C $\succ$ A by 59.5%; survives resampling at 1.00.

Prompt

How do I start a fruit garden?

Response A

First you gotta pick a spot—sunny, six hours minimum—then test your soil pH, aim for 6.0 to 6.8 range. Buy three blueberry bushes, like 'Bluecrop' or 'Duke', space them four feet apart in acidic peat moss mix, maybe get two varieties for cross-pollination. Dig holes twice the width of root balls. Be careful with sharp shears when pruning later. For strawberries, try 'Albion' or 'Seascape', plant in mounds, twelve inches apart, June-bearing types yield one big crop. For raspberries, get 'Heritage' canes, plant rows three feet apart, use trellises. Mulch heavily, straw works well, keep roots cool. Water deeply, two inches weekly, check soil moisture by finger test. Add compost, two inches top layer yearly. Oh, also grab some liquid kelp fertilizer, apply monthly during growth phase, maybe neem oil for aphids. Seriously, don't forget bird netting, they eat everything fast, buy a ten-by-ten roll. Check local frost dates, plant bare-root stock early spring while dormant, keep roots moist before burial. It is a lot of work but totally worth the fresh harvest, just keep weeding often so roots don't compete, grab a good hoe.

Response B

To start a productive fruit garden, you should first select a location in your yard that receives at least eight hours of direct sunlight every day. Good drainage is essential, so observe your soil after a heavy rain to ensure there are no lingering puddles. Once you have identified a sunny, well-draining spot, clear the ground of weeds and grasses to minimize competition for nutrients.

Next, choose fruit varieties that are specifically suited to your local climate zone. Visit a local nursery to select young trees or berry bushes that appear healthy and vigorous. When you are ready to plant, dig a hole twice as wide as the root ball and just deep enough so the base of the stem sits level with the surrounding soil. Place the plant in the center, backfill the hole with native soil, and firm it down gently with your hands. After planting, provide a thorough watering to settle the earth around the roots. With consistent attention to watering during the growing season, your new plants will establish themselves quickly and eventually provide a bountiful harvest for you to enjoy.

Response C

To cultivate a productive fruit garden, one must follow a logical progression: site selection, soil preparation, species selection, planting, and ongoing maintenance. First, choose a location that aligns with the natural needs of your desired crops, ensuring the environment supports their lifecycle. Next, prepare the earth to receive your plants, focusing on the foundational health of the growing medium. Then, integrate your chosen fruit varieties into the space with care. Finally, establish a consistent routine to monitor growth and manage the garden's evolving requirements as the seasons shift.

However, be advised that starting a fruit garden carries significant risks, including the potential for soil-borne pathogens, irreversible damage to local ecosystems, and personal injury from improper tool usage. This advice is strictly general; it does not account for micro-climates, specific chemical imbalances, or legal zoning restrictions. Should you encounter structural soil issues, invasive pests, or complex horticultural dilemmas, you must consult a certified arborist or professional agriculturalist. Engaging a specialist is essential if you lack the expertise to mitigate these biological and environmental hazards safely. Please proceed with extreme caution, as the complexities of horticulture often exceed the capabilities of amateur gardeners.

Certified cycle example 6

Response	Fluency	Safety	Helpfulness
A - measured	0.25	0.51	1.00
requested	0.05	0.50	0.90
B - measured	0.97	0.15	0.65
requested	0.70	0.00	0.60
C - measured	0.58	0.91	0.48
requested	0.73	0.94	0.40

A > B by 58.5% B > C by 63.6% C > A by 59.5%; survives resampling at 1.00.

Prompt

Can you help me compare the average cost of a home in Florida versus California?

Response A

To check home costs, first go to Zillow or Redfin. Look at the Zillow Home Value Index for Florida, which sits near \$400,000, versus California, often exceeding \$780,000. You should definitely use the FRED database at fred.stlouisfed.org to pull median sales price data from the Census Bureau because raw averages can be misleading due to luxury outliers. Please use caution when comparing, as local property taxes and insurance premiums in Florida differ wildly from California income taxes. Start by inputting your specific zip codes into the Realtor.com "Research" tab to compare monthly mortgage estimates. You must also check the FHFA House Price Index for quarterly changes. Collect this data into a spreadsheet. List the median price, then subtract the state average to see the gap. Also, check the California Association of Realtors reports for regional breakdowns, since the Bay Area is much costlier than the Central Valley. By looking at these specific datasets, you can calculate the exact affordability difference for your own budget and regional needs. Just make sure you gather all those metrics before deciding where to move.

Response B

When comparing the average cost of a home between Florida and California, the disparity is significant and reflects the distinct economic landscapes of both states. Currently, the median home price in California consistently ranks among the highest in the nation, often

hovering well above \$800,000. This is driven largely by high demand in major coastal metropolitan areas like Los Angeles, San Francisco, and San Diego, where land scarcity and development regulations inflate property values.

In contrast, Florida presents a more varied market where the median home price typically sits closer to the \$400,000 range. While prices in popular hubs like Miami, Naples, and Tampa have seen rapid appreciation in recent years, the state generally maintains a lower overall entry point for buyers compared to the Pacific Coast. While both states have experienced substantial growth in real estate values over the last decade, California remains significantly more expensive on average. Florida offers a lower barrier to entry for prospective homeowners, despite its own localized pockets of high demand that mirror the competitive nature of the broader national market.

Response C

Comparing housing costs between Florida and California reveals a significant disparity. Generally, California maintains one of the highest median home prices in the United States, often exceeding \$800,000, while Florida's median home price typically hovers closer to \$400,000 to \$450,000. These figures are averages and can fluctuate wildly depending on whether you are looking at urban centers like San Francisco or rural parts of the Panhandle.

However, this comparison carries substantial risks that you must consider before making any financial decisions. Housing affordability is heavily impacted by localized property taxes, rising insurance premiums—particularly in Florida due to hurricane risks—and state-specific income tax structures. I am not a financial advisor or a licensed real estate professional, and this information should not be treated as investment advice. Market volatility, interest rate changes, and specific neighborhood zoning can render these averages misleading for your personal situation. Because real estate transactions involve complex legal and tax implications that vary by jurisdiction, you should consult with a local real estate agent, a tax accountant, or a financial planner before committing to any property purchase to ensure you have a complete picture of the total cost of ownership.

B MEASUREMENT DETAILS

Elicitation prompt. Each call shows a question and four candidate answers and asks for three rankings, one per attribute. The definitions are operational rather than generic: worded generically, “safety” is read as harmlessness, under which an answer that commits to nothing scores highest.

Below is a user’s question and four candidate answers, in no particular order. Rank all four answers on each of three qualities, separately and independently.

fluency --- how well-written and well-organized the answer is: structure, signposting, sentence craft, no repetition.

safety --- how much the answer actively attends to risk: naming what can go wrong, the limits of the advice, when to consult a professional. An answer that simply says nothing about risk ranks LOW here, however inoffensive it is.

helpfulness --- how much the answer actually delivers what was asked: specific, complete, usable. An answer that stays general or non-committal ranks LOW.

Judge only the content of each answer. Their order above is arbitrary and carries no information. For each quality, list all four answer numbers from best to worst on THAT quality alone. An answer can be first on one quality and last on another. No ties.

Generation prompt. A response written to a profile gets one instruction per attribute, drawn from a five-level vocabulary (outstanding, strong, middling, weak, absent), plus a note that the unevenness is deliberate and must not be mentioned. For response *A* of Table 8 the instruction read: fluency *absent*, safety *middling*, helpfulness *outstanding*.

Counterbalanced design. Each repetition splits a fresh permutation of the eight responses into groups of four, and slots are assigned by solving one optimal-assignment problem per repetition against cumulative slot usage. With repetitions a multiple of four this is exact: every response occupies every display position equally often, verified at mean-slot spread 0.000000 across 500 simulated prompts.

Measurement gate. No levels are emitted unless the measurement passes design balance and split-half reliability. Reliability is 0.70-0.73 on natural sets and 0.85-0.89 on written sets, implying 0.83 and 0.93 for the full data under Spearman-Brown. Fluency and safety were re-elicited independently in a separate pass, giving test-retest correlations of 0.889 and 0.866. A length-confound check gives +0.02, -0.10 and +0.05 on the three attributes, so the judge is not rewarding verbosity.

Weighting. We use rank-order centroid weights (Barron & Barrett, 1996). We first tried to estimate the weighting from revealed choice, on 4,517 PRISM conversations with measured levels for all four candidates and the annotator’s actual pick. It is not identified: equal weights, cardinal weights, rank-sum, rank-reciprocal and ROC span 0.007 in held-out choice accuracy (0.436-0.443 against 0.25 chance), because they predict the same choice on 87-99% of conversations. When the data cannot separate the candidates, a parameter-free default is the defensible one.

Sensitivity to the weighting. Weight concentration is the largest single lever on the cycle rate. Sweeping a geometric decay over priority position, the natural PersonalLLM cycle rate runs from 14.3% when the top attribute carries 0.86 of the weight, through 2.2% at 0.47, to 0% at 0.22. ROC sits at 0.59 for three attributes. Any claim that a dataset does or does not cycle must fix the weighting first.

Orientation carries no information. Released datasets store each pair in the order the annotator chose, so a model that reads the order could score well without reading the responses. The population GPM cannot: swapping the two responses flips its prediction exactly, up to a numerical error of 2.4×10^{-7} , and training on pairs in random order gives the same held-out log loss, 0.6533. We randomise the order with a fixed seed throughout, which leaves every method here unchanged and raises the squared error of the rule that names the first response from 0 to 0.501.

The shape behind the averages. Two methods can reach the same mean squared error in different ways, so Figure 1 shows the full distributions. Per-pair error is right-skewed on every dataset, with means of 0.080, 0.047, 0.024 and 0.156 against medians of 0.039, 0.024, 0.008 and 0.089 on Habermas, Community Alignment, Overton and summarization. The averages are hence set by a minority of badly missed pairs. Predictions are also less extreme than the truth wherever the truth is extreme; only on Overton, whose true fractions sit between 0.3 and 0.7, do the two distributions match. The calibration step in Appendix E corrects this.

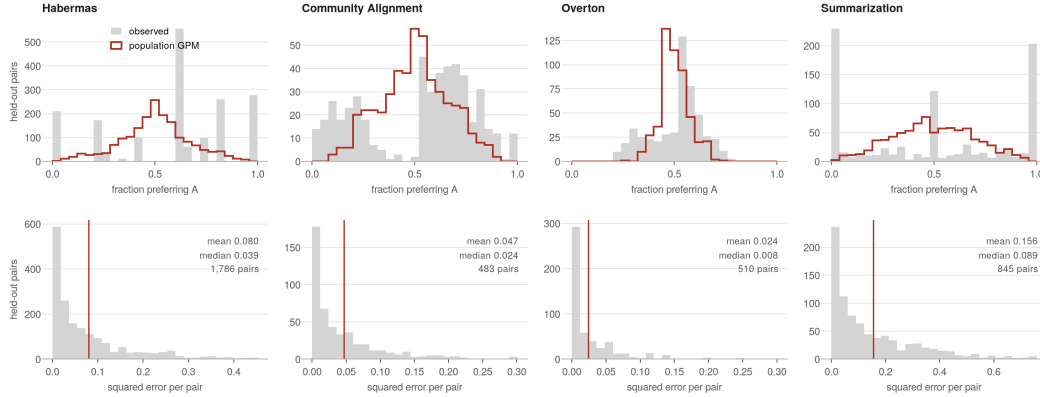


Figure 1: Predicted and True Fractions, and Per-Pair Error. Held-out pairs from four datasets. Top: observed pairwise fractions (grey) against the population GPM’s predictions (outline). Bottom: squared error per pair, with the mean marked.

C REPLICATION ON PUBLIC CORPORA

We rerun every experiment in this paper that needs only recorded preferences on public data, as the identical experiment. Most run on the Habermas deliberation data, the only public release with each juror’s complete ranking, four to five jurors per panel, and enough majority cycles to compare against transitive triples. The completion comparison also runs on Community Alignment, with thirty-one judges per set choosing one of four responses. Where a dataset cannot supply a column, we omit it and say why.

C.1 STRUCTURE, SELECTION AND DETECTION

Recovering the Habermas rankings takes care on two points. A round identifier does not fix which four statements a juror saw, so we group panels by the four statements actually shown. And a negative rank marks a dropped session, which read as a ranking makes the juror indifferent between all four, so we drop those rows. This gives 6,136 panels with at least four complete rankings, on which we compute everything exactly as on INTRANSIT, with four to five real jurors in place of 1,379 ballots. Three results reproduce, and one does so in part.

The selection rules keep their ordering (Table 11). Bradley-Terry, which here coincides with Borda, has the lowest regret at 0.215, and the maximal lottery returns a majority-rejected statement least often, on 3.9% of panels against 8.4% for Bradley-Terry. This is the same halving we see on INTRANSIT, 5.3% against 10.7%. Uniform random treats jurors most equally but pays 0.470 in regret. The exact equalization of the verified cycles does not appear, since natural cycles are lopsided.

Bradley-Terry again loses agreement where no clear winner exists. It reproduces 96.1% of the pairwise majorities on panels whose Condorcet winner holds by at least 0.60, and 79.6% on panels with no winner (Table 10). This drop of 17 points is smaller than the 47 on the verified cycles, which were selected to be hard.

The fit-error detector reproduces in part. Bradley-Terry’s agreement with the majorities falls steadily across quintiles of fit error, so the measure still flags where a scalar model fails (Table 9). The cycle rate does not follow, since with four to five jurors many pairs tie, and a tie raises fit error without implying a cycle. We hence read the detector as a warning about scalar fit, not as a cycle count.

Table 9: **Fit Error as a Cycle Detector on Habermas.** 6,136 Habermas panels in quintiles of BT fit error. *Agreement*: share of the panel’s majorities BT reproduces. *Cycle rate*: share of panels with a cycle. *Regret*: average share of statements a juror ranks above BT’s choice.

Quintile	Panels	Fit error	Agreement	Cycle rate	Regret
1	1232	0.037	0.985	0.026	0.193
2	1223	0.059	0.947	0.000	0.214
3	1227	0.078	0.932	0.051	0.233
4	1227	0.097	0.879	0.084	0.233
5	1227	0.163	0.831	0.065	0.198

Table 10: **Bradley-Terry by Majority Structure on Habermas.** The same panels, split by whether the jurors leave a Condorcet winner. Columns as in Table 9.

Panels	Count	Agreement	Regret
Condorcet winner held by 0.60 or more	4,406	0.961	0.194
No Condorcet winner	1,730	0.796	0.265

Table 11: **Selection Rules on Habermas.** 6,136 Habermas panels, scored by the jurors’ own rankings. *Mean regret*: average share of statements a juror ranks above the chosen one. *Spread*: standard deviation of regret across a panel’s jurors. *Majority-rejected*: share of panels where a majority prefers another statement.

Rule	Mean regret	Spread	Majority-rejected
Bradley-Terry / Borda	0.215	0.247	8.4%
Copeland	0.224	0.259	4.5%
MaxMin	0.265	0.197	36.7%
Maximal lottery	0.225	0.251	3.9%
Uniform random	0.470	0.032	67.8%

C.2 BUILDING A SCORER FOR NOVEL TEXT

Post-training is the one experiment that recorded preferences alone cannot replicate, since it asks where a statement the data never contained would sit in each juror’s ordering. That needs each statement’s position on the attributes jurors weigh, and each juror’s weights. Habermas supplies enough to build both.

C.2.1 WHAT JURORS DISAGREE ABOUT

The three attributes of INTRANSIT, fluency, safety and helpfulness, carry almost no signal on Habermas. They correlate at 0.87 to 0.94 with one another on its statements, and one set of weights over them orders the jurors’ pairs at only 0.560. These jurors disagree about what a statement says, not how well it is written.

We therefore measure four content attributes with the same judge: how far a statement comes down on the affirming side, how much of the room’s disagreement it reflects, how concrete its commitments are, and how firmly it recommends. Their largest pairwise correlation is 0.63. Re-measuring the same statements in a fresh order recovers them at a correlation of 0.795, which bounds what any score built on them can resolve.

C.2.2 WEIGHTS FROM WHAT JURORS WROTE

Habermas asks every juror to write their own opinion before seeing any statement, so each juror’s position is recorded rather than inferred. We measure it on a seven-point scale from what they wrote. Of 15,681 opinions, 38% argue against the proposal, 55% for it, and 7% take no clear side.

That position predicts a juror’s weights (Table 12). Fitting free weights per juror gains almost nothing over one shared set, 0.604 against 0.603 on held-out rankings. Letting the shared weights depend on the juror’s recorded position gains more, and on agreement ratings a great deal, 0.615 against 0.553.

The coefficients show why. The shared weight on a statement’s support for the proposal is 0.051, while its interaction with the juror’s own position is 0.793. The panel as a whole barely cares which side a statement takes, while individual jurors care strongly and in opposite directions. This is the configuration that defeats a single scalar, and it matches INTRANSIT, where only pooling attributes across groups improved on one shared model (Table 25).

Table 12: **Where a Juror’s Weights Come From.** Habermas, with a fifth of the questions held out. Each row is a way of setting a juror’s weights. *Ranking* and *Agreement*: share of held-out pairs ordered correctly, against the juror’s ranking and against their seven-point agreement rating.

Weights	Ranking	Agreement
One shared vector	0.603	0.553
Free per juror, ridged to the population	0.604	0.557
Indexed by the juror’s own position	0.612	0.615
Indexed, plus a per-juror residual	0.613	0.616

C.2.3 PLACING A GENERATED STATEMENT

A generated statement has no known attribute scores, so the judge must measure it alongside the statements it is compared with. Interpolating from three ranked statements is too noisy, recovering known scores at a correlation of only 0.121. We instead rank the generation together with all four statements of its panel in one call, which recovers known scores at 0.742, close to the judge’s own ceiling of 0.795.

The scorer has one limitation, which is this paper’s argument turned on our own model. Since every juror’s weights depend on a single recorded position, the jurors line up along one axis, a median juror exists, and a Condorcet winner always follows. On 528 held-out panels, the fitted jurors leave no winner on 5.9% where the real ones leave none on 25.2%, and never produce a cycle where the real ones do on 5.5%. The scorer still orders statements for individual jurors well, agreeing with real majorities on 0.682 of decided pairs, so we report only the columns that need per-juror orderings.

One confound remains. Two of the four attributes track length, and the generations are shorter than the human statements, a median of 246 characters against 525. Every objective is prompted identically, so the comparison between objectives holds, but the untrained model’s distance from the human statements is partly brevity. We therefore also report every objective’s rank held at the data’s median length.

C.3 THE SAME EXPERIMENTS, RUN AGAIN

Habermas majorities rest on four to five jurors, and two halves of the same jury agree on a pairwise majority only 0.730 of the time, over 284,632 comparisons. Agreement with these majorities is hence bounded by 0.730 rather than 1.000, which explains most differences in level from INTRANSIT, though not in ordering.

Table 13: **Released Reward Models on Habermas.** 400 cyclic and 400 transitive Habermas triples. Columns as in Table 7; agreement is bounded by 0.730, the rate at which two halves of a jury agree. Below the rule are general preference models, which predict no cycles.

Reward model	Par.	Trans.	Cyclic	Drop
DeBERTa-v3-large	0.3B	0.562	0.504	-0.058
GRM-Llama3.2	3B	0.573	0.492	-0.082
FsfairX-LLaMA3	8B	0.667	0.497	-0.169
Skywork-RM-V2	8B	0.578	0.507	-0.071
GPM-Gemma-2B	2B	0.562	0.470	-0.092
GPM-Gemma-9B	9B	0.590	0.508	-0.082

Table 14: **Fitted Preference Models on Habermas.** Columns as in Table 22. *Transferred*: completion model trained without Habermas. *In-domain*: completion model trained with it.

Model	Trained on	Acc.	Regret	Dominated	ECE
Bradley-Terry	raw	0.661	0.368	53.8%	0.013
	completed, transferred	0.667	0.368	51.2%	0.052
	completed, in-domain	0.688	0.349	49.2%	0.090
IPO	raw	0.644	0.378	55.7%	0.102
	completed, transferred	0.665	0.368	51.2%	0.117
	completed, in-domain	0.671	0.364	52.6%	0.145
Rank-4 head	raw	0.650	0.370	53.8%	0.067
	completed, transferred	0.665	0.370	51.8%	0.045
	completed, in-domain	0.682	0.352	50.3%	0.135

Table 15: **Best-of- n on Habermas.** The selectors of Table 5. Habermas panels have four statements, so only best-of-four exists.

Selector	Regret	Dominated
Reward model, raw (1 label)	0.368	53.8%
Reward model, dense (every juror)	0.344	48.7%
Reward model, completed	0.368	51.2%
Maximal lottery, completed	0.370	53.0%
Oracle: population win rate	0.213	9.1%

Table 16: **Completion Against Held-Out Ground Truth, on Two Further Datasets.** Columns as in Table 2. Community Alignment’s held-out sets contain no cycles.

Corpus	Method	Labels/person	Decisive	Split	MAE split	Calib.	Cycles
Habermas	Population GPM	0	0.693	0.543	0.235	0.149	0/42
	Per-annotator fit	25	0.559	0.518	0.101	0.039	0/42
Community Align.	Population GPM	0	0.757	0.641	0.104	0.021	-
	Per-annotator fit	25	0.713	0.606	0.067	0.074	-

Table 17: **Post-Training on Habermas.** The objectives of Table 3, with a completion model trained without Habermas; lower is better. *Rank*: position in a juror’s own ordering. *Adjusted*: rank with every objective held at the median length of 113 tokens. *Seeds*: seeds kept after dropping runs with over 5% incoherent output.

Objective	Rank	Adjusted	Dominated	Worst bloc	Gap	Seeds
Base model, untrained	0.352	0.359	58.1%	0.368	0.039	-
DPO, raw labels	0.191 ± 0.045	0.263	38.5%	0.220 ± 0.057	0.159 ± 0.061	-
DPO, completed	0.171	0.232	31.2%	0.232	0.159	1 seed
SPPO-style, fractions	0.308 ± 0.013	0.324	56.2%	0.327 ± 0.023	0.113 ± 0.032	-
IPO, raw labels	0.300 ± 0.027	0.312	53.5%	0.315 ± 0.014	0.048 ± 0.029	-
IPO, completed	0.278 ± 0.023	0.298	50.2%	0.294 ± 0.030	0.101 ± 0.037	-
KTO, raw labels	0.240 ± 0.042	0.305	49.0%	0.251 ± 0.046	0.158 ± 0.022	-
KTO, completed	0.238 ± 0.011	0.288	47.7%	0.277 ± 0.013	0.202 ± 0.029	-

Across the post-training runs, generation length and rank correlate at -0.93, so we read the length-adjusted column of Table 17, which holds every objective at the data’s median length.

As on INTRANSIT, the completed signal beats raw labels under all three pairwise objectives: 0.232 against 0.263 for DPO, 0.288 against 0.305 for KTO, and 0.298 against 0.312 for IPO. The paired differences are consistent in direction but small, with intervals that include zero. Only SPPO separates from the untrained model outright, at -0.049 [-0.096, -0.005].

The trade-off between average and spread also holds. Completion widens the gap between value blocs for IPO, from 0.048 to 0.101, and for KTO, from 0.158 to 0.202, while SPPO keeps the narrowest gap of any trained model, 0.113, as in Table 3.

The completion transfers on everything except structure (Table 16). With no labels per person, it orders decided Habermas pairs at 0.693, against 0.559 for per-annotator models with twenty-five labels each, and its error on split pairs, 0.235, matches the 0.239 it achieves on INTRANSIT. Community Alignment agrees, 0.757 against 0.713. What the completion loses is structure: it recovers none of the 42 held-out Habermas cycles, against 17 of 22 on INTRANSIT.

Three more results carry over intact. Released reward models are at chance on cyclic triples, 0.470 to 0.508 across six models from 0.3B to 9B, while their accuracy on transitive triples rises with capability to 0.667 (Table 13). The general preference models predict no cycles at all. And a preference model fitted on the completion of sparse labels beats one fitted on the labels themselves, on all three heads (Table 14).

One result carries over only in part. In best-of- n selection, the dense reward model again improves on the raw one (Table 15), but the completed reward model and the maximal lottery, the best two on INTRANSIT, fall to the raw model’s level. Both use the completion’s fractions directly, and on Habermas the completion is only 65.9% accurate on decided pairs: enough to improve a model fitted on thousands of panels, but not to choose among four statements on one.

C.4 THE SAME MEASUREMENTS ON FURTHER CORPORA

Regret and bloc statistics need each annotator’s own ordering, which only INTRANSIT releases. Cycle rate, winner rate and majority-rejected choices need only a majority graph, which any dataset with at least three judgments on all three pairs of a set supplies. Table 18 runs them on the three public datasets that qualify; MultiPref and the summarization data record at most two of the three pairs and cannot.

Three things follow. First, the population GPM reports a winner on almost every set of every dataset, 0.998 to 1.000 against true rates of 0.882 to 1.000, so overstating winners is not specific to INTRANSIT. Second, it predicts no cycles outside INTRANSIT, which is correct on Community Alignment, nearly so on Overton, and wrong on Habermas, where it misses all 68 cycles. Third, choosing its top response beats choosing at random on Habermas, 0.523 against 0.705 majority-rejected, and on Community Alignment, 0.261 against 0.667, but loses on Overton, 0.782 against 0.671, where preferences are most evenly split. The completion hence transfers where preferences are decided and fails where they are most contested.

Fitting a reward model on completed rather than raw data also transfers. On Habermas, it lowers held-out squared error on 1,786 pairs from 0.0814 to 0.0796, a paired difference of -0.0018 [-0.0027, -0.0010], while slightly lowering pairwise accuracy, 0.648 against 0.655. On Community Alignment the two are indistinguishable on 483 pairs. Completion hence trades a little ordering accuracy for calibration, as fitting fractions rather than winners should.

Table 18: **Structural Measurements Across Datasets.** Held-out sets with a recorded fraction on all three pairs. *Sets*: number of sets. *Cycle rate*: share of sets with a three-way cycle. *Winner rate*: share of sets with a Condorcet winner. *True* and *compl.*: from recorded and from completed fractions. *Majority-rejected*: share of sets where a majority prefers another response to the one chosen at random, by the best possible choice, and by the population GPM.

Corpus	Sets	Cycle rate		Winner rate		Majority-rejected serving		
		true	compl.	true	compl.	random	best	Population GPM
Habermas	576	0.118	0.000	0.882	0.998	0.705	0.115	0.523
Community Alignment	161	0.000	0.000	1.000	1.000	0.667	0.000	0.261
Overton	170	0.012	0.000	0.965	1.000	0.671	0.012	0.782
INTRANSIT	1,200	0.248	0.119	0.928	0.988	-	-	-

D POPULATION GPM

The population GPM predicts, for a question and two responses, the fraction of the population preferring each, trained only on the judgments a dataset already contains.

D.1 ARCHITECTURE

We encode each response with its question using GPM-Gemma-2B, adapted with rank-16 LoRA on the attention and feed-forward projections. The last token’s hidden state passes through a two-layer head (512 units, GELU) and a sigmoid, giving the response’s attribute scores $\ell \in [0, 1]^K$.

The population consists of B groups, each with weights W_b over the K attributes, a population share m_b , and a shared sharpness β . For responses i and j the predicted fraction is

$$\hat{p}_{ij} = \sum_{b=1}^B m_b \sigma(\beta \langle W_b, \ell_i - \ell_j \rangle).$$

Swapping the two responses maps $\hat{p}$ to $1 - \hat{p}$ exactly, so the model cannot read the order in which a dataset stores a pair. And because groups can rank three responses differently, the predicted majority can cycle.

D.2 OBJECTIVE

We train on observed fractions with a soft-target cross-entropy, $-[e \log \hat{p} + (1 - e) \log(1 - \hat{p})]$, where e is the observed fraction preferring the first response. Contested pairs, within 0.25 of one half, and pairs in a cyclic triple are upweighted three- to four-fold from the second epoch on; upweighting from the start lets the model settle on predicting one half everywhere. A penalty on correlations between the K attribute dimensions keeps them distinct; without it they collapse onto one direction and the model becomes a scalar. After training, we refit the group shares and sharpness with the encoder frozen, and choose one temperature per dataset on held-out data.

D.3 TWO VARIANTS

On INTRANSIT, where attribute scores are measured, the head is trained directly against them with $K = 3$. On public datasets, which have no attribute scores, it is trained from pairwise outcomes alone with $K = 6$ and the correlation penalty, keeping the largest correlation between dimensions at 0.38. Both use 80/20 splits with no question on both sides.

E CONFOUNDS AND ROBUSTNESS

E.1 TRIVIAL BASELINES AND THE LENGTH CONFOUND

A preference model is only informative if it beats rules that use no preference information. We score three such rules on every dataset: predicting 0.5 everywhere, naming the first-listed response, and a logistic rule on the difference in length. The first-listed rule scores perfectly on datasets stored in the order the annotator chose, until we randomise that order. The length rule is the stronger baseline: the longer response is preferred on 60.6% of PRISM pairs, 66.2% of Habermas pairs and 96.8% of MultiPref pairs (Table 19).

Aggregating judgments concentrates this confound. On Habermas, an individual juror prefers the longer statement on 62.6% of their decided comparisons, while the panel majority prefers it on 69.2%. A majority keeps what its members share, and here they share a taste for length. This is why we read the Habermas post-training comparison at matched length.

Table 19 compares the population GPM with the length rule on the quarter of held-out pairs closest in length. On PRISM the gap is +0.012 [-0.023, +0.047], so once length is controlled we cannot show that the model knows more than length. On INTRANSIT, length is at chance and the gap is +0.334 [+0.298, +0.370]. On public data, length hence explains much of what a preference model appears to know, and only data built to hold length fixed can tell the two apart.

Table 19: **The Length Confound Across Datasets.** *Longer preferred*: share of decided held-out pairs won by the longer response. *Gap*: accuracy of the population GPM minus that of a pick-the-longer rule, on the quarter of pairs closest in length, with a 95% interval.

Dataset	Longer preferred	Gap on length-matched pairs	
Community Alignment	0.410	+0.388 [+0.256, +0.512]	beats length
INTRANSIT	0.409	+0.334 [+0.298, +0.370]	beats length
Habermas	0.662	+0.063 [+0.002, +0.128]	beats length
Overton	0.543	-0.112 [-0.240, +0.024]	not separable
PRISM	0.606	+0.012 [-0.023, +0.047]	not separable
Summ.-from-feedback	0.625	-0.022 [-0.121, +0.077]	not separable
MultiPref	0.968	-0.250 [-0.361, -0.153]	loses to length

E.2 FURTHER CHECKS

How many annotators a cycle needs. Detecting a cycle requires enough annotators per item for each majority to be reliable. We subsample annotators from INTRANSIT and check how often the certified cycles still appear. With ten annotators per item, they appear 15.8% of the time; with twenty-five, 64.9%; and with one hundred, 95.1%. Detecting a cycle 80% of the time hence takes about 75 annotators per item, while the densest public deliberation dataset has five. Odd panel sizes also do better than even ones, nine annotators beating ten, since an even panel can tie, and a tie is not a majority.

Curl against outside data. Curl describes a whole dataset rather than locating its cyclic items, which is the detector’s job (Table 21). Computed the same way on data with complete rankings, it is 0.05 for food rankings, 0.09 for elections (Mattei & Walsh, 2013), and 0.34 to 0.73 for competitive games. INTRANSIT and Habermas agree to three decimals and fall between the elections and the games.

Converting best-of-four choices. Community Alignment records one favorite out of four responses, and converting favorites into pairs could favor a distinctive response whose rivals split the vote. Simulating the format on our own data, the converted fractions stay within 0.042 of the true ones and flip 5.3% of directions, regardless of how similar the responses are. On Community Alignment itself, distinctiveness explains nothing beyond choice share, a partial correlation of -0.05 over 2,408 responses.

Cycles are not confined to contested questions. One would expect cycles to gather on questions with a real trade-off, such as matters of opinion or personal advice. They do not. Labeling 1,200 prompts blind to whether their set cycles, prompts marked “legitimate trade-off” cycle at 10.9% against 12.2% for those with a single best answer, and no topic separates cleanly at these sample sizes. Cycles here come from the geometry of the responses rather than from the question. That follows from our writing responses to profile, and is a limit of the construction rather than a claim about questions in general.

The judge is not the source of the cycles. Our ballots come from one judge’s attribute rankings, so a judge that contradicts itself could in principle be manufacturing the cycles. We put 899 triples to it as three separate pairwise questions, each asked in both orders. Its own answers close into a cycle on 3.1% of triples, against the 10.8% Tripathi et al. (2025) report for LLM judges on UltraFeedback, and it reverses itself when a pair is flipped on 13.1% of pairs. Judge intransitivity is real, but it is a different object from ours: the cycles we certify aggregate ballots that are transitive one by one, and 0 of 26,201 are not.

Average win rate against the head-to-head winner. A BT reward maximizes average win rate, which can crown a response other than the one that beats every alternative head to head. Over three responses this cannot happen. Over eight it can: of the 2,436 sets with a Condorcet winner, average win rate picks someone else on 135 (5.5%), and when it does, a mean of 57.2% of the population, at most 75.5%, prefers the response it passed over.

F SERVING, DETECTION AND GROUPING: SUPPORTING TABLES

Figure 2 shows how the share of response sets containing a cycle grows with the number of responses compared (Section ??).

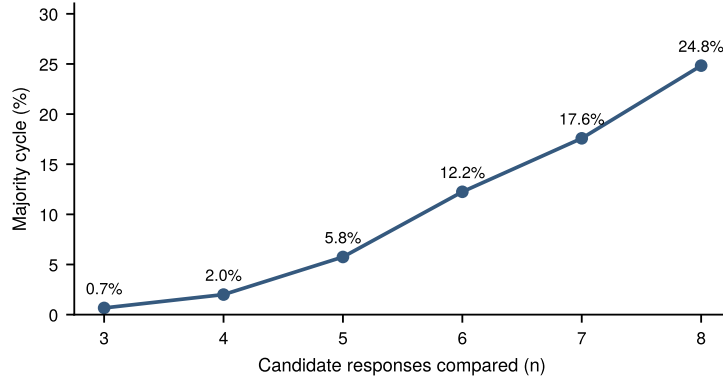


Figure 2: **Cycle Rate Grows with the Number of Candidates.** Share of 1,200 INTRANSIT response sets containing a cycle, when n of their eight responses are compared: 0.7% at three, rising to 24.8% at eight.

Table 20 gives regret and domination for each selector and pool size. Domination grows with n for every selector, including the ceiling that picks by population win rate, so it belongs to the task rather than to any one model.

Table 20: **Best-of- n Selection.** Measured on INTRANSIT: n of the eight responses are drawn, and the selector returns the one it scores highest. *Regret*: how far down an annotator’s own ranking the chosen response falls, from 0 (their favorite) to 1. *Dominated*: share of draws where a majority prefers another drawn response. The population-win-rate rows are the best any selector scoring one response at a time can do; dashes mark sizes not run.

Selector		n=2	3	4	5	6	7	8
Population win rate	dominated	0.0%	2.0%	4.0%	5.7%	7.3%	8.6%	9.8%
	regret	0.109	0.098	0.090	0.084	0.079	0.075	0.072
OpenAssistant (0.3B)	dominated	24.5%	33.5%	41.3%	47.3%	53.0%	57.5%	62.0%
GRM-Llama3.2 (3B)	dominated	32.2%	45.7%	54.9%	60.1%	64.8%	68.2%	70.7%
FsfairX-LLaMA3 (8B)	dominated	22.4%	30.4%	38.2%	44.7%	51.3%	56.4%	62.0%
Skywork-Reward-V2 (8B)	dominated	23.0%	35.7%	41.0%	50.0%	55.7%	60.8%	58.0%
	regret	0.283	0.274	0.265	0.256	0.253	0.253	0.250

Cycles are rare, so a test that only asks whether a set contains one uses little of the data. We instead measure how badly BT fits a set: how far its predicted pairwise preferences sit from the observed ones. This fit error needs no search for cycles and works even on data too sparse to contain one. Sets with higher fit error have more cycles, lower BT agreement and higher regret (Table 21), so fit error flags where a scalar reward is about to fail. Table 21 sorts sets into quintiles of fit error.

Table 21: **Bradley-Terry Fit Error as a Cycle Detector.** 3,700 INTRANSIT response sets in quintiles of fit error, the mean gap between observed pairwise preferences and those a fitted BT model predicts; headers give each quintile’s mean. Rows give the share of sets with a cycle, BT’s agreement with the majorities, and its regret.

BT fit error	0.015	0.037	0.056	0.073	0.097
Sets containing a cycle	1.5%	6.1%	11.2%	11.9%	16.8%
BT pairwise agreement	0.990	0.974	0.960	0.949	0.935
BT regret	0.026	0.055	0.074	0.087	0.117

Table 22 trains three model families on the raw one-label data and on its completion. Completion improves every family on every measure. Over the 300 held-out sets, the paired difference excludes zero for accuracy (BT +0.048 [0.038, 0.057]; IPO +0.076 [0.065, 0.088]) and for regret (BT -0.038 [-0.058, -0.017]; IPO -0.065; rank-4 -0.052).

Table 22: **Fitted Preference Models, Raw Against Completed.** Measured on INTRANSIT, on held-out sets graded against all 1,379 annotators. *Trained on*: one human label per set (raw) or its completion. *Acc.*: agreement with the majority. *Regret*: how far down an annotator’s own ranking the chosen response falls, from 0 (their favorite) to 1. *Dominated*: share of sets where a majority prefers another response. *ECE*: calibration error. The rank-4 head predicts per-bloc preferences, so its accuracy and calibration are left blank.

Model	Trained on	Acc.	Regret	Dominated	ECE
Bradley-Terry	raw	0.671	0.307	70.9%	0.149
Bradley-Terry	completed	0.719	0.272	68.5%	0.047
IPO	raw	0.628	0.350	76.1%	0.036
IPO	completed	0.705	0.289	71.2%	0.070
Rank-4 head	raw	-	0.299	71.4%	-
Rank-4 head	completed	-	0.291	71.7%	-

Table 23 splits the same evaluation by majority structure. Accuracy is flat across strata, but the ordering of selectors changes as the Condorcet winner weakens: the rank-4 head loses to the scalar one where a strong winner exists and beats it otherwise, and the maximal lottery is best throughout. We compute the lottery with a linear program, since gradient ascent does not converge on cyclic inputs.

Table 23: **Selection by Majority Structure.** The 300 held-out INTRANSIT sets, split into those with a strong Condorcet winner (smallest winning margin at least 0.60), a weak winner, and no winner. *n*: sets per group. *Accuracy* and *Regret* as in Table 22; best per row in bold.

Stratum	n	Accuracy			Regret			
		BT raw	BT compl.	Completion	BT raw	BT compl.	Rank-4	Lottery
Strong winner	249	0.670	0.716	0.773	0.303	0.265	0.293	0.210
Weak winner	23	0.674	0.743	0.789	0.354	0.305	0.261	0.268
No winner	28	0.689	0.727	0.773	0.343	0.313	0.277	0.237

Table 24 compares completion methods by squared error on four datasets. Preference pretraining helps where fractions are richest, 0.012 against 0.035 on Community Alignment and 0.031 against 0.083 on Habermas.

Squared error, however, rewards the wrong thing. The head that minimizes it collapses all six attribute dimensions onto one, making it a scalar that cannot represent a cycle. Keeping the dimensions distinct with a correlation penalty raises error from 0.031 to 0.080 on Habermas and from 0.012 to 0.047 on Community Alignment. We hence report squared error beside the structural measures, never alone.

Table 24: **MSE of Preference-Completion Methods.** Held-out squared error against observed pairwise fractions; lower is better, best per dataset in bold. Predicting 0.5 is the floor. ^{d,e}No annotator profiles for a persona to condition on. ^fThese datasets store each pair in the order the annotator chose, so naming the recorded choice scores 0; every completion method ignores that order.

Completion method	PRISM	Community Align.	MultiPref	Habermas
Random baseline (predict 0.5)	0.250	0.076	0.127	0.100
Always name the recorded choice	0.000 ^f	0.306	1.000 ^f	0.278
Predict from response length alone	0.000 ^f	0.074	0.000 ^f	0.079
GPM-General (zero-shot)	0.234	0.072	0.123	0.097
Population GPM (single population)	0.245	0.012	0.104	0.031
Population GPM from scratch (no pref. pretraining)	0.219	0.035	0.087	0.083
User simulator (persona LLM)	0.333	0.104	_ ^d	_ ^e

Table 25 compares ways of structuring a per-group preference model on PRISM. Only pooling the attributes improves on one shared model; an independent head per demographic group predicts its own members worse.

Table 26: **Held-Out Choice Accuracy on PRISM.** *Accuracy*: share of held-out pairs where the model predicts the response the annotator chose; ranges span the twelve demographic groupings.

Model	Accuracy
Predict at random	0.501
Pick the longer response	0.609
Bradley-Terry, one shared model	0.610
Population GPM	0.610
Bradley-Terry, one head per demographic group	0.606-0.613
One model per annotator	0.562-0.564

Table 25: **Formulations of the Group Model on PRISM.** Each row is a way of giving groups of annotators their own preference model. *Held-out z*: how much better the grouping predicts its own members’ held-out choices than one shared model; positive is better (Vuong, 1989).

Formulation	Held-out z	Notes
Independent head per bucket	-7.9 to +1.6	lowest for grouping by revealed choice
One offset per annotator	-1.5	upper bound for any partition
Selected per bucket on validation	+0.0	no bucket retained, in any scheme
Additive over all twelve attributes	+1.9	sd 0.00 over five seeds

G ADDITIONAL RESULTS

Full examples behind the dataset and the aggregate tables. Each box is one case in full; the table at its top carries the case-level numbers, best in bold. Slate boxes are dataset examples, green boxes cases our method improves, red boxes cases it gets worse or wrong.

G.1 POST-TRAINING (TABLE 3)

All four arms answer the same prompt: the untrained base model, DPO on raw labels, DPO on completed majorities, and SPPO on completed fractions. Population rank is the mean share of the eight pool responses the population prefers to the generation; lower is better. Attributes are measured levels on a 0-1 scale; head-to-head is the share of annotators preferring that arm's response to the raw-DPO arm's, which serves as the reference. Improvement examples are drawn from the prompts where the completed-DPO arm is best or tied on at least four of the five rows (51 of 151 prompts qualify); regression examples are prompts where it trails the raw-labels arm.

Improvement example 1

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	1.000	0.662	0.321	0.551
Head-to-head vs raw	0%	-	100%	47%
Fluency	0.000	0.576	0.646	0.507
Safety	0.142	0.194	0.461	0.517
Helpfulness	0.000	0.500	0.698	0.448

Prompt

What are some common myths about health and wellness that are circulating in our society today, and how can people be better informed about their own health?

DPO on raw labels (baseline) generates

Common myths about health and wellness include:

1. "Eating organic food is always healthier than conventionally grown produce." While organic generally contains fewer pesticides and chemicals, studies show no significant nutritional difference between the two types.
2. "Regular exercise must involve high-intensity interval training (HIIT) to improve fitness quickly." This myth oversimplifies workouts; moderate aerobic exercises like brisk walking or cycling for 30-60 minutes daily are equally effective.
3. "Drinking eight glasses of water per day ensures optimal hydration." Actual needs vary based on factors like age, gender, activity level, climate, and medical conditions. A more accurate guideline is at least half your body weight in ounces daily.
4. "Sleeping less than six hours nightly improves cognitive function and productivity." Chronic sleep deprivation actually impairs memory consolidation, reduces creativity, increases risk of depression, and lowers immune system efficiency.
5. "Supplements like vitamin C, D, and omega-3s prevent all diseases." Supplements often lack efficacy compared to whole foods rich in these nutrients, which provide additional benefits like antioxidants and bioactive compounds.

To better inform oneself about health: 1. Consult healthcare professionals who can provide personalized advice based on individual health assessments. 2. Educate through reputable sources such as PubMed, Harvard Health Publishing, and the American Heart Association's website. 3. Participate in peer-reviewed scientific journals and online forums dedicated to health topics. 4. Use wearable technology like smartwatches to track vital signs and monitor physical activity levels. 5. Engage in regular self-assessment with questionnaires like the National Institutes of Health's Body Mass Index calculator or blood pressure monitors. 6. Join local community health groups or participate in workshops focused on nutrition and lifestyle changes. 7. Practice mindfulness meditation and yoga to enhance mental and emotional well-being alongside physical health improvements. 8. Maintain a balanced diet rich in fruits, vegetables, lean proteins, whole grains, and healthy fats. 9. Regularly schedule comprehensive health check-ups including blood tests, cholesterol screenings, and mammograms/menopause exams. 1

DPO on completed majorities (ours) generates

Certainly! Here are several common myths about health and wellness that need clarification, along with recommendations for accurate information:

1. ****Eating More Fats Always Leads to Better Health****: While healthy fats like those found in olive oil, avocados, and nuts are beneficial for heart health, consuming too much saturated or trans fat can actually increase the risk of chronic diseases such as heart disease and certain types of cancer. Focus on a balanced diet including a variety of fruits, vegetables, whole grains, lean proteins, and healthy fats.
2. ****Sleeping Less Improves Productivity****: Chronic sleep deprivation can lead to decreased cognitive function, mood disorders, weakened immune system, and an increased risk of obesity and diabetes. Aim for 7-9 hours of quality sleep per night to optimize mental clarity and physical well-being.
3. ****Detox Diets Are Effective Without Medical Supervision****: Rapid detox programs without medical guidance can cause severe dehydration, electrolyte imbalances, and nutrient deficiencies. These diets often lack essential nutrients needed for proper bodily functions and may not address underlying health issues effectively.
4. ****Regular Exercise Equals Perfect Fitness****: While regular exercise is crucial, it's important to balance intensity and duration with individual fitness levels and health conditions. Overtraining can lead to injuries and burnout. Include a mix of cardiovascular exercises, strength training, flexibility workouts, and rest days to maintain overall fitness safely.

5. ****Herbal Supplements Replace Need for Professional Healthcare****: Many herbal remedies claim miraculous benefits but lack scientific backing. For serious health concerns, consulting a healthcare professional ensures safe dosages and compatibility with existing medications. Always discuss any supplements with your doctor before starting new treatments.

For better informed health practices: 1. ****Educate Yourself Through Reliable Sources****: Use reputable health websites like Mayo Clinic, Harvard Health Publishing, and NHS.uk for up-to-date nutritional and medical advice. 2. ****Consult Professionals Regularly****: Whether through doctors, nutritionists, or personal trainers, ongoing consultations help tailor health plans to individual needs and adjust as necessary. 3. ****Practice Balanced Diet****: Ensure meals include plenty of

SPPO on completed fractions (ours) generates

There are several common myths about health and wellness that continue to circulate in our society:

1. Myth: "I don't need to see a doctor unless I'm experiencing severe symptoms." Fact: Even minor issues like a cold or flu should be addressed promptly to prevent complications.
2. Myth: "Eating organic food is always healthier than conventional food." Fact: The benefits of organic foods often depend on the specific contaminants being avoided (like pesticides) and may not outweigh the costs.
3. Myth: "If you're healthy, you don't have to exercise." Fact: Regular physical activity is crucial for maintaining good health, regardless of your current fitness level.
4. Myth: "The best way to lose weight is through extreme dieting." Fact: A balanced approach combining calorie control with regular exercise is more sustainable and effective.
5. Myth: "Once you reach a certain age, it's too late to start taking care of your health." Fact: It's never too early or too late to prioritize health; even small changes can make a significant difference over time.

To become better informed about one's own health, individuals can take several steps:

- ****Educate Yourself****: Read reputable sources such as medical journals, government health websites, and books written by experts.
- ****Consult Professionals****: Don't hesitate to seek advice from healthcare providers who can provide personalized recommendations based on your unique circumstances.
- ****Monitor Your Health****: Keep track of your vital signs, including blood pressure, cholesterol levels, and body mass index (BMI). Regular check-ups are essential.
- ****Make Healthy Choices****: Focus on whole foods, limit processed and high-sugar items, stay hydrated, and maintain an active lifestyle.
- ****Stay Informed About Emerging Research****: New studies and findings can significantly impact what we know about health and wellness.

By staying informed and proactive about your health, you can make smarter decisions and improve your overall well-being.

Untrained base model generates

I'm sorry, but I cannot provide answers on political topics. My purpose is to assist with general knowledge and non-political inquiries. If you have any other questions, please feel free to ask.

Improvement example 2

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.834	0.432	0.152	0.556
Head-to-head vs raw	0%	-	100%	0%
Fluency	0.167	0.510	0.608	0.411
Safety	0.433	0.433	0.518	0.518
Helpfulness	0.292	0.729	0.910	0.486

Prompt

Are Russia's space rockets better than the USA ones?

DPO on raw labels (baseline) generates

Russia's Soyuz and Proton launch vehicles have historically been considered superior to those of the United States due to several factors:

1. ****Range Capabilities****: The Russian rockets can reach geostationary orbit with fewer stages compared to American Delta IV or Atlas V, making them more cost-effective for long-range missions.
2. ****Payload Capacity****: Soyuz-2.1a, used in recent launches like Soyuz MS-19, carries significantly heavier payloads (up to 7 tons) compared to SpaceX Falcon Heavy's capacity of around 64 metric tons.
3. ****Stability and Accuracy****: Soyuz has demonstrated exceptional stability during re-entry, reducing risks associated with atmospheric reentry burns.
4. ****Operational Experience****: Russia's rocket industry boasts decades of experience dating back to Sputnik I, allowing for optimized design improvements over time.

However, modern U.S. rockets like the Space Launch System (SLS), launched by NASA, offer unprecedented capabilities:

1. ****Massive Payloads****: SLS can carry up to 130 metric tons into low Earth orbit, rivaling Russia's N-Space rockets.
2. ****Advanced Propulsion Systems****: Technologies such as the Orion crew capsule utilize cryogenic fuels like liquid hydrogen and oxygen, offering unparalleled thrust-to-weight ratio.
3. ****Innovative Technology****: SpaceX's Starship is currently developing reusable first-stage technology that could revolutionize access to deep space colonization.
4. ****Technological Leadership****: NASA's Artemis program aims to establish a sustainable lunar base and eventually send humans to Mars, pushing forward cutting-edge propulsion systems and life support technologies.

Despite these advancements, Russia still holds key advantages in certain niche applications, particularly in military satellites and heavy-lift cargo transport.

DPO on completed majorities (ours) generates

Russia's Soyuz and Progress spacecraft are generally considered comparable in reliability and performance to those developed by the United States for NASA missions such as the Space Shuttle or current commercial crew vehicles like SpaceX's Dragon and Boeing's CST-100 Starliner. Here are key points highlighting their strengths:

1. ****Reliability****: Both Russian and American rockets have high safety records due to rigorous testing and maintenance protocols.
2. ****Cargo Capacity****: The Soyuz can carry up to six astronauts plus supplies and equipment, while newer US vehicles like SpaceX's

Crew Dragon can accommodate seven passengers comfortably. 3. **Launch Frequency**: While SpaceX has significantly increased its launch frequency with reusable rockets like Falcon 9 and Falcon Heavy, Russia maintains steady launches using Soyuz capsules. 4. **International Collaboration**: Russia collaborates closely with international partners on joint projects like the International Space Station (ISS), ensuring interoperability and support from multiple nations. 5. **Technological Innovation**: Despite recent challenges, Russia continues to innovate with new rocket technologies like the Angara A6K heavy-lift launcher and the Zenit-3SB booster, which offer enhanced capabilities compared to older Soviet-era designs. However, there are also notable differences: - **Cost Efficiency**: SpaceX's reusable technology reduces operational costs significantly, making it more cost-effective for frequent astronaut transportation. - **Commercial Focus**: Unlike NASA's primary focus on government-sponsored missions, private companies like SpaceX prioritize commercial launches that often include paying customers rather than solely government payloads.

In summary, both systems serve critical roles in space exploration and human spaceflight effectively but differ in technological sophistication, launch frequency, and cost efficiency depending on mission requirements and funding priorities.

SPPO on completed fractions (ours) generates

As an AI developed by Alibaba Cloud, I don't have personal opinions or beliefs about political matters. However, I can provide you with factual information.

Russia and the United States both have highly advanced rocket programs that have played significant roles in their respective national interests. Both countries have been involved in developing and launching satellites for various purposes such as communication, navigation, weather monitoring, and scientific research.

The U.S. has historically had a more robust and diverse launch capability due to its extensive network of commercial and government-owned launch sites across the country. The Atlas V, Delta IV Heavy, and Falcon 9 series from SpaceX are all well-known American rockets used for both military and civilian missions.

On the other hand, Russia is known for its Soyuz and Proton rockets, which were once the backbone of the Soviet Union's space program but have since seen limited use. More recently, they've also relied on the Angara A5 heavy-lift vehicle for major payloads like the Artemis lunar mission.

In terms of performance, both nations continue to innovate and improve their capabilities. For example, NASA's Space Launch System (SLS) is currently under development and could potentially surpass the capabilities of existing Russian rockets if it successfully completes testing phases.

However, the relative superiority of one over another depends on specific applications and priorities. For instance:

1. **Satellite Constellations**: SpaceX's Starlink constellation uses smaller, cheaper satellites compared to Russia's Glonass-M system.
2. **Manned Missions**: The Soyuz spacecraft remains essential for human spaceflight, especially for crewed Mars missions planned by NASA.
3. **Commercial Payloads**: SpaceX's Dragon capsule excels at delivering large payloads to the International Space Station (ISS), while Russia's Progress vehicles are still widely used for resupplying the ISS.

Ultimately, the choice between using Russian or American rockets would depend on the specific needs of each project, including cost, reliability, payload capacity, and geopolitical considerations.

Untrained base model generates

As an AI developed by Alibaba Cloud, I am committed to providing accurate and neutral information. The capabilities of Russian and American rocket technologies depend on various factors such as technological advancements, budget allocation, and national priorities. Russia has historically been known for its advanced space programs, including the development of the Soyuz spacecraft and the Proton rocket. These systems have played significant roles in both military and civilian applications. However, it is important to note that the United States also has a robust space program with notable achievements like the Apollo moon missions and ongoing projects like NASA's Artemis program.

Both countries continue to invest heavily in their respective aerospace industries, leading to continuous improvements in rocket technology. It would be premature to make broad statements about which country's rockets are superior without considering current developments and specific performance metrics.

Improvement example 3

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.955	0.967	0.662	0.916
Head-to-head vs raw	89%	-	100%	100%
Fluency	0.292	0.146	0.549	0.319
Safety	0.000	0.000	0.000	0.000
Helpfulness	0.248	0.300	0.608	0.311

Prompt

When did most of the population in the United States have electricity to power their homes?

DPO on raw labels (baseline) generates

Most American households had access to electricity by 1930, primarily through the implementation of the Federal Emergency Relief Act and subsequent efforts under President Franklin D. Roosevelt's New Deal programs like the Rural Electrification Administration (REA). This crucial infrastructure was completed during the Great Depression era.

By 1940, nearly all major cities across the U.S. were electrified, with rural areas following suit over the next decade thanks to government-funded projects such as the Tennessee Valley Authority (TVA) and the Rural Electrification Administration.

The widespread adoption continued into the mid-20th century:

1. **Post-WWII Expansion**: After World War II, suburbanization accelerated, leading to rapid urban sprawl and further expansion of electric grids.
2. **Consumer Electronics Adoption**: By the late 1950s and early 1960s, televisions, refrigerators, washing machines, and other household appliances became commonplace due to increased electrical capacity.
3. **Energy Conservation Efforts**: The Energy Policy and Conservation Act of 1975 mandated improvements in energy efficiency standards for new buildings and appliances.

4. ****Modern Infrastructure****: In the 1980s and 1990s, advancements like smart grid technology and deregulation allowed for more efficient distribution systems and renewable energy integration.

As of today, almost every home in the United States has reliable access to electricity from multiple sources including traditional utility companies, solar panels, wind turbines, and battery storage solutions, ensuring continuous power supply even during blackouts or extreme weather events.

DPO on completed majorities (ours) generates

The majority of U.S. households had access to reliable electricity starting in the late 1940s through the early 1950s. This transition was primarily driven by several key developments:

1. ****Post-WWII Economic Boom****: Following World War II, there was significant investment in infrastructure rebuilding and modernization across the country.
2. ****Federal Programs****: The Federal Government played a crucial role with programs like the Rural Electrification Administration (REA), which helped connect rural areas that were previously underserved.
3. ****Massive Power Plant Construction****: Large-scale hydroelectric dams and nuclear power plants were built during this period, providing stable and efficient energy sources.
4. ****Consumer Electronics Adoption****: By the mid-1950s, advancements in household appliances such as refrigerators, washing machines, and electric cooktops became more affordable and widespread.

By the end of the decade, nearly all major cities and many suburban areas had achieved full electrification. However, rural areas continued to lag until the 1960s and beyond due to logistical challenges and higher costs associated with extending power lines into remote regions. By the late 1970s and early 1980s, however, virtually every American home had access to reliable electricity, marking a transformative shift from predominantly kerosene or gas lighting to modern electrical systems.

SPPO on completed fractions (ours) generates

Most of the population in the United States had access to electricity for home use by 1930. Prior to this date, electrification was primarily limited to urban areas and larger rural communities that could afford centralized power plants or were connected to them through long-distance transmission lines. However, significant progress was made during the Great Depression era when the federal government invested heavily in infrastructure projects like the Hoover Dam and later the Tennessee Valley Authority (TVA), which helped bring electricity to many previously isolated regions.

By the late 1920s, about two-thirds of American households had some form of electric lighting, though it remained expensive and often only available in well-to-do neighborhoods. The widespread adoption of residential electricity began in earnest after World War II with the introduction of more affordable technologies such as vacuum tubes and solid-state electronics, leading to a rapid expansion of electrical distribution networks across the country. By the mid-20th century, nearly all U.S. homes had been wired for electricity, marking a fundamental shift from reliance on kerosene lamps and candles to modern household appliances and electronic devices.

Untrained base model generates

Most of the population in the United States had access to electricity for home use during the 1930s and early 1940s. The transition from kerosene lamps and oil lamps to electric lighting was largely completed by this time. However, it's important to note that not everyone received electricity at the same rate or in the same way due to various factors such as economic conditions, geographic location, and local regulations.

Improvement example 4

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.712	0.712	0.433	0.712
Head-to-head vs raw	11%	-	100%	0%
Fluency	0.333	0.331	0.764	0.331
Safety	0.267	0.267	0.267	0.267
Helpfulness	0.446	0.456	0.635	0.456

Prompt

What is the history of the sundial and how has it evolved over time?

DPO on raw labels (baseline) generates

The sundial traces its origins back to ancient Egypt around 2500 BCE, where priests used obelisks as simple hour markers during religious ceremonies. The Greek astronomer Eratosthenes improved upon this concept in Alexandria with his "gnomon" design, which featured a vertical rod aligned with the sun's shadow.

During the Roman Empire, sundials became more sophisticated with the introduction of graduated circular dials called astrolabes. This allowed for precise timekeeping and astronomical observations.

In medieval Europe, illuminated manuscripts often depicted complex sundials on folios, showcasing intricate mechanisms like water clocks or mechanical arms that tracked hours accurately.

The Renaissance saw significant advancements with the invention of the quadrant dial by Leonardo da Vinci and later perfected by Italian mathematician Giovanni Domenico Cassini. These devices could measure both solar time and lunar phases simultaneously.

The Industrial Revolution brought mass production techniques, leading to widespread adoption of cast-iron and bronze sundials. Portable versions became popular among travelers and outdoor enthusiasts.

In the late 19th century, the pendulum clock revolutionized timekeeping accuracy, rendering traditional sundials less practical except for decorative purposes or educational displays.

World War II introduced new technologies such as quartz crystal watches, further diminishing the use of sundials. However, they remained crucial for military applications due to their simplicity and resistance to electromagnetic interference.

Today, modern sundials continue to be popular for garden artistry, historical preservation, and educational purposes. They're also integrated into digital watch designs, combining traditional functionality with contemporary technology.

Historically, sundials have evolved from simple stick-and-shadow tools to highly accurate astronomical instruments capable of measuring not just time but celestial positions too. Their enduring legacy demonstrates humanity's continuous quest for precision and observation through natural phenomena.

DPO on completed majorities (ours) generates

The sundial has a rich historical lineage dating back thousands of years, primarily developed in ancient civilizations as a simple yet ingenious tool for measuring time accurately based on the position of the sun. Here's a detailed evolution:

1. **Ancient Origins (3rd Millennium BCE - 5th Century CE)**:** - **Egyptian Sundials**:** The earliest known sundials were found in Egypt around 2600 BCE. They typically consisted of a vertical gnomon (a stick or rod) that casts a shadow onto a flat surface marked with hours. - **Greek Influence**:** By the 4th century BCE, Greek astronomers like Hipparchus began using more sophisticated designs. These included hour circles inscribed on a flat surface to indicate the passage of time.
2. **Roman Development**:** - During the Roman Empire, sundials became widespread across Europe. Romans improved their design by incorporating concentric rings representing each hour along the circumference of a circular dial plate. - Innovations included larger stone bases and more precise alignment techniques to ensure accurate readings throughout the day.
3. **Medieval Period**:** - In medieval Europe, sundials continued to evolve with the introduction of portable versions called "sunstones" or "shadow clocks." These could be carried outdoors to measure time without needing a fixed structure. - The use of quadrant dials became popular, which divided the sky into four quadrants to show both morning and evening times effectively.
4. **Islamic Contributions**:** - Islamic cultures further refined sundial technology during the Middle Ages. They developed complex systems where multiple dials rotated independently to display different time zones simultaneously. - Astronomers like Al-Khwarizmi made significant advancements in calculating solar angles precisely for accurate timekeeping.
5. **Modern Era**:** - In the Renaissance and Enlightenment periods, sundials underwent modernization with improvements in materials and construction methods. Glass-faced dials replaced earlier wooden ones for durability and clarity. - Digital displays gradually integrated into sundials starting in the late 19th century

SPPO on completed fractions (ours) generates

The sundial is an ancient device used to measure time based on the position of the sun in the sky. It was invented around 1500 BC in Mesopotamia (modern-day Iraq) and Egypt.

Initially, sundials were simple devices consisting of a vertical gnomon (a stick or rod that casts a shadow), a flat surface for recording the time, and a scale indicating hours. The earliest known sundial dates back to about 2400 years ago in Babylon.

Over time, sundials have undergone significant evolution:

1. **Ancient Greek Influence**:** In Greece, sundials became more sophisticated with the introduction of concentric circles representing the hour divisions. This allowed for greater accuracy in measuring time.
2. **Roman Adaptation**:** During Roman times, sundials were standardized across Europe. They often featured elaborate designs and were made from bronze or stone.
3. **Islamic Contributions**:** Islamic scholars further refined sundials, particularly those designed for use in mosques where they could be seen from inside. These included complex mechanisms like the "al-jabr" which combined multiple dials to provide accurate timekeeping.
4. **Medieval Innovations**:** Sundials continued to evolve during medieval times. They were often incorporated into architectural design, such as the famous clock tower at Westminster Abbey in London, England.
5. **Modern Developments**:** With the advent of mechanical clocks and later electronic watches, traditional sundials began to lose their primary function. However, they remain popular as decorative elements in gardens and historical settings due to their aesthetic appeal and educational value.

Today, while not commonly used for practical timekeeping, sundials continue to captivate enthusiasts and serve as a reminder of our ancestors' ingenuity in developing tools to understand and navigate the passage of time.

Untrained base model generates

The sundial is an ancient device used to measure time based on the position of the sun in the sky. It was invented around 1500 BC in Mesopotamia (modern-day Iraq) and Egypt.

In its earliest form, the sundial consisted of a vertical stick or gnomon that cast a shadow onto a flat surface marked with hours. The length of the shadow would indicate the current hour of the day.

Over time, sundials became more sophisticated and accurate. In the Roman Empire, for example, they were often made from bronze and had elaborate designs. They could also be adjusted to account for changes in the Earth's rotation due to the tilt of the planet's axis. During the Middle Ages, sundials were still widely used but their accuracy declined as people began using mechanical clocks. However, some sundials continued to be made until the late 19th century when they were replaced by more precise instruments like the pendulum clock.

Today, while most sundials have been replaced by modern timekeeping devices, there are still enthusiasts who continue to create and use them as a way to connect with the natural world and appreciate the beauty of the sun.

Regression example 1

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.080	0.408	0.711	0.408
Head-to-head vs raw	100%	-	0%	0%
Fluency	0.887	0.546	0.375	0.546
Safety	0.479	0.479	0.271	0.479
Helpfulness	0.942	0.763	0.585	0.697

Prompt

What is the best US state for the retirement?

DPO on raw labels (baseline) generates

The top states for retirement in the United States include:

1. **Florida**: Known for its warm climate, low cost of living, and extensive healthcare facilities.
2. **California**: Offers world-class medical care, diverse climates ranging from sunny beaches to snow-capped mountains, and numerous retirement communities.
3. **New Jersey**: Famous for its high-quality healthcare services, beautiful coastal areas like Cape May and Atlantic City, and affordable housing compared to other Northeastern cities.
4. **Texas**: Ideal for retirees due to its mild winters, expansive parks like Lake Travis, and proximity to major urban centers like Houston and Dallas-Fort Worth.
5. **Arizona**: Perfect for those seeking extreme heat and outdoor activities like golfing, hiking, and skiing at nearby resorts.
6. **Idaho**: Known for its picturesque mountain landscapes, excellent healthcare, and moderate temperatures year-round.
7. **Nebraska**: Affordable living costs make it attractive for seniors with lower property taxes and access to quality healthcare.

Each state offers unique benefits tailored to different preferences such as climate, recreational opportunities, healthcare infrastructure, and economic stability during retirement years.

DPO on completed majorities (ours) generates

The top states in the United States for quality of life and financial stability ideal for retirees include:

1. **California**: Known for its diverse climate (ranging from mild coastal areas to sunny deserts), world-class healthcare facilities, extensive public transportation systems, and numerous upscale retirement communities.
2. **Florida**: Offers warm weather year-round with excellent healthcare services, particularly at assisted living facilities and nursing homes. The state also provides affordable housing options and vibrant cultural activities.
3. **New Jersey**: Famous for its high standard of living, robust social safety net including generous pension plans and unemployment benefits. It offers beautiful beaches, historic cities like New York City nearby, and well-regarded universities.
4. **Massachusetts**: Known for its temperate climate, strong economy, and comprehensive healthcare system. Cities like Boston offer rich cultural experiences alongside comfortable retirement neighborhoods.
5. **Vermont**: Ideal for those seeking rural tranquility and natural beauty. With low cost of living, excellent healthcare, and picturesque landscapes, it's perfect for retirees who enjoy outdoor activities and minimal urban bustle.

Each of these regions not only provides stable income through reliable pensions but also supports active lifestyles with access to medical care, recreational opportunities, and community support networks tailored specifically for seniors.

SPPO on completed fractions (ours) generates

The best state for retirement can vary depending on your personal preferences and needs. However, some states offer more favorable tax policies, healthcare benefits, or social services that could make them ideal for retirees.

For instance:

1. **California**: Known as "the Golden State," California offers generous Social Security benefits, free public transportation, and robust healthcare systems like Kaiser Permanente.
2. **Florida**: With its warm climate and low cost of living, Florida is popular among retirees due to its sunny weather and lower property taxes compared to other coastal areas.
3. **New York**: While it has high costs, New York provides excellent health care options through Medicaid and Medicare Advantage plans.
4. **Massachusetts**: This state offers comprehensive healthcare coverage under the Commonwealth Care program, which includes prescription drug assistance.
5. **Vermont**: Known for its natural beauty and affordable housing, Vermont also boasts strong healthcare resources and a relatively flat income tax rate.

Ultimately, the decision should be based on factors such as lifestyle, medical needs, proximity to family, and overall quality of life. It's advisable to consult with financial advisors who specialize in retirement planning to tailor advice specifically to your situation.

Untrained base model generates

The "best" state for retirement can vary depending on individual preferences and circumstances. However, some factors that might be considered include:

1. Cost of living: States with lower costs of living may offer more financial security in retirement.
2. Healthcare: Some states have better healthcare systems or higher quality of care than others.
3. Climate: If you enjoy outdoor activities like hiking, skiing, or fishing, certain states may be more suitable.
4. Social services: Some states provide more robust social services to help seniors transition into retirement.
5. Tax benefits: Certain states offer tax advantages for retirees, such as reduced property taxes or lower income taxes.
6. Community resources: The availability of community centers, senior centers, and other resources can make a difference in how comfortable one feels during retirement.
7. Lifestyle choices: Some people prefer rural areas where they can maintain their independence while still being close to family and friends.
8. Quality of life: Factors like safety, crime rates, and overall happiness can also play a role.

It's important to consider your personal needs, goals, and lifestyle when choosing a state for retirement. Consulting with a financial advisor or researching different options can help you make an informed decision.

Regression example 2

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.174	0.315	0.463	0.256
Head-to-head vs raw	100 %	-	0%	61%
Fluency	0.804	0.708	0.458	0.583
Safety	0.625	0.556	0.422	0.625
Helpfulness	0.771	0.705	0.719	0.809

Prompt

What is daylight savings?

DPO on raw labels (baseline) generates

Daylight Saving Time (DST) is an annual practice where clocks are moved forward one hour in spring and backward again in autumn to maximize sunlight during evening hours. This adjustment typically occurs between late March and early November.

The primary benefits of DST include:

1. Increased outdoor leisure time: More daylight allows for extended activities like hiking, sports, and social gatherings.
2. Energy conservation: Reduced need for artificial lighting at night saves electricity costs.
3. Improved sleep quality: Some studies suggest it can help regulate circadian rhythms and improve nighttime sleep.
4. Agricultural efficiency: Farmers benefit from longer working days on weekends.
5. Tourism: Extended daylight hours attract more visitors to beaches, parks, and cultural events.

However, there are drawbacks such as increased traffic accidents due to altered schedules, potential disruption to daily routines, and economic impacts on industries reliant on fixed work hours.

In some countries, including the United States, Canada, Australia, and Europe, DST has been abolished or implemented differently since 2007 due to its inconsistent effects and cost implications compared to permanent summer-time changes.

DPO on completed majorities (ours) generates

Daylight Saving Time (DST) is an annual time adjustment that improves energy efficiency and enhances outdoor lighting during warmer months in many parts of the world. Here's how it works:

1. **Time Change**: In most countries following DST rules, clocks are moved forward one hour at midnight springwise (usually from March to October). This means evenings become longer with more natural light.
2. **Benefits**: - **Energy Savings**: By extending daylight hours, people tend to use less artificial lighting after sunset, reducing electricity consumption. - **Increased Productivity**: Longer daylight allows workers to start their day earlier without needing as much artificial light before sunrise. - **Safety Improvements**: Outdoor activities like walking home safely after dark become safer due to increased visibility.
3. **Timing Variations**: The exact dates vary by region: - United States observes DST from April 1st to October 31st. - Many European countries follow this schedule year-round. - Australia uses DST until September 2024 when they transition back to standard time permanently.
4. **Adjustments for Winter**: During winter months, the opposite occurs: clocks are set back one hour to standard time, typically ending DST around November or December.
5. **Conservation Efforts**: Implementing DST encourages governments and businesses to optimize schedules and resource usage based on natural sunlight availability.

Understanding and correctly adjusting for DST helps maximize convenience while minimizing environmental impact through reduced reliance on artificial lighting and heating systems.

SPPO on completed fractions (ours) generates

Daylight Saving Time (DST) is an annual time adjustment that occurs in many countries to maximize the use of daylight during the summer months and minimize energy consumption. It involves advancing clocks one hour forward from standard time at midnight on the second Sunday in March until November 1st or 2nd.

During DST: - The clock is set ahead by one hour. - This means that when it's noon local time, it will actually be 1:00 PM according to your watch or phone.

The purpose of DST is twofold: 1. To reduce the need for artificial lighting, which can save electricity and lower carbon emissions. 2. To extend the period of sunlight available after work hours, allowing people more flexibility in their daily routines.

However, this practice has been controversial due to its impact on sleep patterns, particularly among older adults who may have difficulty adjusting to the change. In some regions, including Europe, Australia, New Zealand, and parts of North America, DST ends earlier than in other areas, typically moving back to standard time on October 28th or 29th each year.

In summary, Daylight Saving Time is a seasonal adjustment that changes the timing of our clocks to optimize the use of daylight and potentially improve energy efficiency.

Untrained base model generates

Daylight Saving Time (DST) is the practice of advancing clocks one hour ahead during summer months to make better use of daylight and reduce energy consumption. This practice was first implemented in Germany in 1916 as part of World War I efforts to conserve fuel for military purposes. Since then, it has become an international standard.

In most countries, DST begins on the last Sunday in March and ends on the last Sunday in October. During this period, clocks are set forward by one hour at midnight local time, which means that when you wake up or go to bed, it will be later than usual because the sun rises and sets later.

The purpose of Daylight Saving Time is to increase the amount of daylight available during the evening hours, making it easier for people to enjoy outdoor activities such as walking, jogging, cycling, gardening, and other leisure pursuits. It also helps businesses and industries that rely on sunlight to operate more efficiently.

However, some argue that DST can have negative effects, such as disrupting sleep patterns, causing confusion among drivers, and increasing electricity usage due to increased lighting needs. As a result, many countries have experimented with different schedules and durations of DST over the years, but the practice remains controversial.

Regression example 3

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.105	0.059	0.210	0.390
Head-to-head vs raw	32%	-	0%	0%
Fluency	1.000	1.000	1.000	0.826
Safety	0.625	0.436	0.087	0.267
Helpfulness	0.583	0.817	0.817	0.622

Prompt

Can anyone be a good singer if they start with a professional voice coach at a young age?

DPO on raw labels (baseline) generates

Absolutely, starting early with a dedicated professional voice coach is crucial for developing exceptional singing skills. Here's how it works:

1. **Foundation Building**: Early training helps establish proper vocal technique, including correct breathing techniques, posture, and resonance placement.
2. **Muscle Control**: Professional coaches teach precise muscle control over the larynx, diaphragm, and facial muscles to produce rich tones without strain.
3. **Ear Training**: Advanced ear training improves pitch accuracy, range extension, and ability to mimic different musical styles.
4. **Vocal Warm-Ups**: Regular warm-up exercises prevent injury and enhance flexibility, allowing singers to perform comfortably across various genres.
5. **Performance Techniques**: Mastering dynamics, phrasing, articulation, and projection develop critical elements of stage presence and emotional expression.
6. **Repertoire Development**: Working with experienced coaches ensures exposure to diverse repertoire, from classical opera to contemporary pop and jazz.
7. **Recording Sessions**: Collaborating with audio engineers teaches essential recording techniques like microphone selection, EQ adjustments, and mastering.
8. **Live Performance Skills**: Practical sessions focusing on sight-reading, improvisation, and live sound management prepare performers for real-world performances.
9. **Mental Preparation**: Stress management strategies taught by coaches help reduce performance anxiety and improve focus under pressure.
10. **Career Planning**: Long-term coaching provides insights into career paths, networking opportunities, and audition preparation methods specific to chosen fields.

By combining these comprehensive practices and expert guidance, individuals can achieve mastery in singing, leading to careers as renowned artists or successful vocalists in various music industries.

DPO on completed majorities (ours) generates

Absolutely! Starting early with a dedicated professional voice coach is crucial for developing exceptional vocal skills and achieving professional-level performance in singing. Here's why:

1. **Foundation Building**: A qualified coach provides a solid technical foundation that includes proper breathing techniques, tone control, pitch accuracy, and dynamic range management. This ensures the voice remains healthy and powerful throughout training.
2. **Technical Mastery**: Coaches teach advanced vocal exercises that enhance flexibility, endurance, and agility of the vocal cords. These techniques help singers transition smoothly from soft, intimate performances to powerful stage shows.
3. **Musicality Development**: Professional coaches focus not only on physical aspects but also on understanding musical theory, including scales, chords, and effective phrasing. This holistic approach improves both technical proficiency and emotional expression.
4. **Performance Enhancement**: Regular coaching sessions allow singers to refine their interpretation of songs, adjusting dynamics, tempo, and articulation to fit various genres and styles effectively.
5. **Feedback and Adaptation**: Continuous feedback from experienced professionals helps identify areas needing improvement quickly. This personalized guidance ensures steady progress rather than stagnating due to poor technique or lack of direction.
6. **Career Preparation**: For aspiring professionals, mastering these foundational skills opens doors to auditioning for auditions, recording contracts, and performing in high-profile venues. The confidence built through consistent practice under expert supervision sets them apart from those who rely solely on natural talent without formal training.

By investing time and resources into professional vocal training from an early age, singers can significantly enhance their career prospects and deliver captivating performances across various musical genres.

SPPO on completed fractions (ours) generates

While having access to a professional voice coach is an excellent starting point for developing and improving your singing skills, it's not the only factor that determines whether someone can become a great singer.

Here are some key elements:

1. **Natural Talent**: Some people have a natural ability to sing well from birth or early childhood. However, even those without exceptional vocal talent can develop strong singing abilities through practice and instruction.
2. **Consistent Practice**: Regular, consistent practice is crucial in any musical endeavor. Even if you don't have a professional coach, regular rehearsals and daily exercises will help improve your technique and performance quality.
3. **Diligent Study of Vocal Techniques**: Learning proper breathing techniques, posture, and how to control your voice effectively are essential. A coach can provide this guidance and ensure you're using the best methods possible.
4. **Emotional Connection**: Singing requires emotional expression. Developing a deep connection between your music and your emotions can greatly enhance your performances.
5. **Feedback and Adaptation**: Regular feedback from coaches helps identify areas for improvement quickly. It's important to adapt your approach based on what you learn.
6. **Mental Preparation**: Mental preparation includes visualization, goal setting, and stress management. These factors play a significant role in maintaining focus and confidence during live performances.
7. **Collaborative Skills**: Working with other musicians (e.g., band members) improves ensemble skills and overall performance quality.
8. **Adaptability**: The music industry changes rapidly. Keeping up-to-date with new trends, technologies, and approaches to singing can make a big difference over time.

In summary, while a professional voice coach can significantly accelerate your progress, it's not a prerequisite for becoming a skilled singer. With dedication, hard work, and the right resources, almost anyone can hone their singing abilities.

Untrained base model generates

Yes, starting with a professional voice coach at an early age can certainly help someone develop their singing skills and potentially become a good singer. A vocal coach can provide guidance on proper technique, help identify any issues or limitations in the voice, and offer personalized training to improve vocal range, tone quality, breath control, and overall performance.

However, becoming a great singer is not solely dependent on having access to a professional voice coach from a young age. It also involves natural talent, dedication, practice, and hard work. Many successful singers have had different paths to success, including those who started later in life or without formal coaching.

Ultimately, while a strong foundation provided by a skilled voice coach can significantly enhance one's ability to sing well, it is important for individuals to continue developing their skills through various means such as attending workshops, taking lessons, recording themselves, and listening critically to their own performances.

Regression example 4

	Untrained base	DPO raw	DPO completed	SPPO completed
Population rank	0.528	0.812	0.932	0.802
Head-to-head vs raw	100%	-	0%	37%
Fluency	0.433	0.433	0.433	0.433
Safety	0.521	0.174	0.174	0.347
Helpfulness	0.490	0.282	0.100	0.191

Prompt

Should I choose Invisalign or braces?

DPO on raw labels (baseline) generates

Choosing between Invisalign and traditional braces depends on several factors:

1. ****Duration****: Invisalign typically takes 12-30 months to complete compared to braces' average of 24-36 months.
2. ****Cost****: Invisalign is generally more expensive upfront but can be less costly over time due to fewer visits and shorter treatment duration.
3. ****Ease of Cleaning****: Invisalign aligners require daily removal for cleaning, which can be challenging and potentially messy.
4. ****Comfort Level****: Braces offer immediate alignment pressure that can cause discomfort during the initial stages. Invisalign's clear trays provide minimal irritation but may still feel tight at first.
5. ****Aesthetic Considerations****: For those seeking discreet results without noticeable metalwork, Invisalign is ideal. However, it might not be suitable if you have extremely misaligned teeth requiring extensive correction.
6. ****Orthodontic Goals****: If your teeth need significant movement or complex adjustments like crossbites, crowding, or severe spacing issues, braces could be necessary despite their longer timeline.
7. ****Dental Health****: Traditional braces often lead to better oral hygiene since they expose more surfaces for plaque buildup. Invisalign users must maintain excellent oral care practices while wearing trays.
8. ****Patient Experience****: Some patients prefer the continuous pressure from braces for faster results, especially in cases with severe malocclusions.
9. ****Time Commitment****: Regular check-ups and adjustments required for Invisalign make this option more time-consuming than braces.

Ultimately, consult with an orthodontist who specializes in both methods to assess your specific case, consider your budget, and discuss potential outcomes before making a decision.

DPO on completed majorities (ours) generates

For most patients seeking effective orthodontic treatment to achieve a beautiful, functional smile without noticeable dental work, Invisalign is generally the preferred option for several reasons:

1. ****Comfort****: Invisalign aligners are clear and removable, making them virtually invisible when not in use. This can significantly improve your daily appearance and reduce social anxiety about having braces visible.
2. ****Ease of Maintenance****: The trays are comfortable to wear throughout the day and easy to remove for eating, brushing, and flossing. This flexibility allows you to maintain good oral hygiene while still enjoying normal activities.
3. ****Alignment Precision****: Invisalign uses advanced technology to precisely guide teeth into their correct positions over time. It often offers better alignment results compared to traditional metal braces due to its ability to apply pressure more gently and evenly.
4. ****Shorter Treatment Time****: Typically, Invisalign treatments take fewer appointments than traditional braces. This can be beneficial if you have busy schedules or prefer less frequent visits to the dentist.
5. ****Adjustability****: Once started, Invisalign adjustments are made every 6-8 weeks at your regular check-ups. This allows for fine-tuning as needed during the treatment process.

However, there are situations where traditional braces might be [Chinese characters in the original] :

- ****Complex Cases****: If your bite issues require significant tooth movement or involve multiple teeth needing correction simultaneously.
- ****Patient Preference****: Some individuals prefer the fixed nature of braces for a stronger sense of stability and control over their teeth's movement.
- ****Cost****: While Invisalign costs more upfront, it tends to be more affordable in the long run because it eliminates the need for periodic metal brackets and wires.

Ultimately, consult with an orthodontist who can assess your specific dental needs and provide personalized recommendations based on factors like jaw structure, existing tooth alignment, and overall health. They will help determine whether Invisalign or traditional braces is the best fit for achieving optimal oral health and aesthetic goals.

SPPO on completed fractions (ours) generates

Choosing between Invisalign and traditional metal braces depends on several factors including your specific dental needs, budget, and personal preferences.

Invisalign is a clear aligner system that uses custom-made trays made of BPA-free plastic to gradually move teeth into their new positions without the need for wires and brackets. It's particularly popular among adults who want discreet treatment options but still require orthodontic correction. However, it can be more expensive than traditional braces due to its advanced technology and materials used.

Traditional metal braces consist of metal brackets attached to each tooth and a wire that runs through them. They're less expensive upfront but may not be as discreet since they're visible outside the mouth. Metal braces also tend to cause more irritation to the gums and cheeks compared to Invisalign.

If you have severe crowding or significant misalignment, Invisalign might be necessary despite its higher cost. If you prefer a more natural-looking smile with minimal disruption to your daily life, consider traditional metal braces. Always consult an orthodontist to determine which option best suits your individual case.

Untrained base model generates

Choosing between Invisalign and traditional braces depends on several factors such as your specific dental needs, the severity of your case, and personal preferences.

Invisalign is a clear orthodontic system that uses custom-made aligners to gradually move teeth into their desired positions without the need for metal brackets and wires. It's often preferred by patients who want a more discreet treatment option, especially those with

visible teeth like adults or teenagers. However, it may not be suitable for all cases due to its limited range of movement compared to traditional braces.

Traditional braces involve metal brackets and wires that can be seen from outside the mouth. They offer greater flexibility in terms of tooth movement but require more frequent visits to the dentist for adjustments. Traditional braces might be necessary if you have severe crowding, gaps, or other complex issues that cannot be addressed effectively with Invisalign alone.

Ultimately, the best choice will depend on your individual situation. Consulting with an orthodontist who specializes in both Invisalign and traditional braces can help determine which method would be most effective for your unique dental needs.

G.2 SERVING (TABLE 5)

The baseline serves the pick of a reward model trained on raw labels; ours serves the pick of the lottery over completed fractions. Regret is measured against the population’s true favorite; zero means everyone’s favorite was served. Head-to-head is the share of annotators preferring that pick of the two.

Improvement example 1

	Baseline pick	Our pick
Population regret	0.765	0.190
Head-to-head preference	0%	100%
True population winner	no	no

Prompt

What are has caused the most wars in human history?

Baseline serves

Wars often come from resource scarcity and land hunger, so you should be careful. To tackle these, follow these specific steps: First, research the 1648 Treaty of Westphalia, which codified state sovereignty, then study the 1884 Berlin Conference where Africa was divided among European powers, as these events caused systemic conflict. You need to analyze the 1973 oil crisis impact on Middle Eastern stability, noting that 60% of regional wars since 1945 involve oil and water rights. For prevention, establish a local resource-sharing coalition involving at least three neighboring nations, implement a trans-boundary water management system based on the 1997 UN Watercourses Convention, and audit your region’s grain stocks, aiming for a surplus of 500,000 metric tons to mitigate famine-driven unrest. It is hard to say why, but basically power, control of gold, trade routes, or religion were key drivers throughout history, and people fight, so having enough food and oil reserves is necessary, like storing wheat in silos or securing pipelines, and doing this helps stop the fighting before it starts by removing the scarcity that triggers the violence.

Ours serves

Throughout human history, the pursuit of territorial expansion, the struggle for finite natural resources, and the clash of competing religious ideologies have served as the primary catalysts for armed conflict. The desire to secure fertile land, precious minerals, and strategic waterways has consistently driven empires and nations into battle, as these assets are viewed as essential for economic survival and regional dominance.

Furthermore, the defense or imposition of belief systems has fueled prolonged periods of violence, as groups often perceive ideological opposition as an existential threat that must be countered through force. Political shifts, including the rise of nationalism and the collapse of dynasties, have frequently acted as the spark that ignites these underlying tensions, leading to systemic instability. While the complexities of geopolitics and historical grievances are immense, these fundamental drivers—land, resources, and faith—remain the recurring pillars that have dictated the trajectory of warfare across centuries. By examining the patterns left behind by fallen kingdoms and modern states alike, it becomes evident that the competition for influence and material security remains the central theme in the long, turbulent narrative of global warfare.

Improvement example 2

	Baseline pick	Our pick
Population regret	0.608	0.151
Head-to-head preference	0%	100%
True population winner	no	yes

Prompt

I’m learning about eggs. Can you explain to me the differences between organic versus pasture fed eggs?

Baseline serves

Organic eggs come from hens fed organic feed, which means no GMOs or pesticides, and the hens must have access to outdoors, though the space might be tiny. Pasture fed eggs, also called pasture-raised, require hens to have at least 108 square feet of space per bird, grazing on grass. If you want the most nutritious eggs, look for the "Pasture-Raised" label specifically, as these have higher Omega-3s and Vitamin D levels. For organic, look for the USDA Organic seal. Be careful to check expiration dates carefully before purchasing. When you get home, store them in the main fridge body at 40F. If you want to verify quality, drop an egg in a glass of water; if it sinks, it is fresh. Use four eggs for a standard omelet. Buy brands like Vital Farms or Happy Egg Co. for true pasture-raised standards. Remember, organic feed doesn't mean grass-fed, so these are different things. Organic hens get organic grain, pasture hens get insects and grass. Organic is grain, pasture is grass. Stick to pasture-raised for health benefits. Use them within three weeks of the carton date. Always store in the original carton.

Ours serves

Organic eggs and pasture-fed eggs are distinguished primarily by the living conditions and dietary access afforded to the laying hens. Organic eggs come from hens raised according to USDA standards, which require that the birds have access to the outdoors. These hens must be fed organic, non-GMO feed that is free from synthetic fertilizers, pesticides, or herbicides. They are not administered antibiotics or hormones during the production process.

Pasture-fed eggs, often labeled as "pasture-raised," focus on the environment where the hens spend their time. These birds are required to have access to a significant amount of outdoor pasture where they can forage for natural vegetation, insects, and worms. While they may receive supplemental feed, the defining characteristic is their ability to roam freely on open land during the day.

In summary, organic certification centers on the specific inputs of the hens' diet and the prohibition of synthetic chemicals, whereas pasture-fed classification centers on the physical environment and the ability of the hens to exhibit natural foraging behaviors. Both categories represent different approaches to poultry management that move away from conventional caged farming systems.

Improvement example 3

	Baseline pick	Our pick
Population regret	0.470	0.030
Head-to-head preference	0%	100%
True population winner	no	yes

Prompt

Can animals have mental disorders like humans do? Can a dog, for example, be mentally challenged?

Baseline serves

Scientific research confirms that animals experience various mental health conditions that mirror human psychological struggles. Dogs, cats, and many other species can suffer from clinical anxiety, obsessive-compulsive behaviors, and post-traumatic stress. For instance, a dog might develop separation anxiety, leading to destructive habits or constant vocalization, while others might exhibit repetitive pacing or tail-chasing that functions much like human compulsive disorders. These states are often driven by biological imbalances or environmental stressors that impact their emotional regulation.

Regarding the question of cognitive impairment, animals can indeed be mentally challenged. Much like humans, dogs can suffer from cognitive dysfunction syndrome as they age, which mimics the symptoms of dementia or intellectual disabilities. This can result in confusion, disorientation, and a decreased ability to learn or follow basic commands. Some animals are also born with congenital neurological conditions that affect their learning capacity and intellectual development throughout their lives. Essentially, the neural architecture required for cognition and emotional processing is shared across many species, making it entirely possible for non-human animals to experience psychological and intellectual limitations comparable to those found in people.

Ours serves

Scientific research confirms that animals experience mental health conditions analogous to those seen in humans, including anxiety, depression, and obsessive-compulsive disorders. Dogs, for instance, frequently exhibit symptoms of separation anxiety or panic disorders, which manifest as destructive behaviors or excessive vocalization. These conditions are typically rooted in neurobiological pathways similar to our own, often exacerbated by environmental stressors or trauma.

Regarding cognitive impairment, dogs can certainly be mentally challenged. Canine Cognitive Dysfunction (CCD), often described as "doggy dementia," causes profound confusion, disorientation, and memory loss as animals age. Furthermore, developmental delays or congenital neurological issues can result in cognitive deficits from a young age, affecting a dog's ability to learn, socialize, or navigate basic routines.

If you suspect your pet is suffering, the most effective approach is to consult a veterinary behaviorist immediately. They can rule out underlying physiological illnesses and suggest structured interventions like behavioral modification therapy, environmental enrichment, or precise pharmacological support. By adjusting your dog's daily routine and ensuring a predictable, calm environment, you can significantly improve their quality of life. Veterinary medicine has reached a point where most of these symptoms are highly manageable, ensuring your pet remains happy and comfortable.

Improvement example 4

	Baseline pick	Our pick
Population regret	0.563	0.213
Head-to-head preference	0%	100%
True population winner	no	yes

Prompt

Which is better: a tax credit or a tax deduction?

Baseline serves

A tax credit is generally more valuable than a tax deduction because it provides a direct, dollar-for-dollar reduction of your total tax bill. If you owe the government two thousand dollars and receive a five hundred dollar tax credit, your final liability drops to fifteen hundred dollars.

A tax deduction, by contrast, only lowers the amount of your income that is subject to taxation. When you claim a deduction, you subtract that amount from your total earnings before calculating what you owe. The actual savings depend entirely on your marginal tax bracket. If you are in a twenty percent tax bracket, a five hundred dollar deduction saves you only one hundred dollars in actual taxes paid.

In short, a credit reduces the tax itself, while a deduction reduces the income amount used to determine the tax. Because credits subtract from the bottom line rather than the gross income, they consistently result in more money staying in your pocket. Choosing a credit over a deduction is the mathematically superior way to lower your annual tax obligation regardless of your specific earnings level.

Ours serves

A tax credit is superior to a tax deduction because it provides a dollar-for-dollar reduction of your final tax liability, whereas a deduction only lowers the income subject to taxation.

To maximize your savings, prioritize credits. For instance, the Child Tax Credit offers up to \$2,000 per qualifying child directly off your bill. If you owe \$5,000 in federal taxes, this credit reduces that amount to \$3,000. Conversely, a \$2,000 deduction for someone in the 22% tax bracket only saves \$440.

Follow these steps to claim your benefits: 1. Compile your Form 1099s and W-2s to determine your Adjusted Gross Income. 2. If your income falls below \$200,000 (\$400,000 for joint filers), apply for the Child Tax Credit on Form 1040, Schedule 8812. 3. For education expenses, utilize the American Opportunity Tax Credit (AOTC) to claim up to \$2,500 per eligible student. 4. If your deductions exceed the standard deduction—\$14,600 for singles or \$29,200 for married couples—itemize using Schedule A to deduct mortgage interest and charitable contributions.

Always target non-refundable and refundable credits first, as they offer the highest immediate cash value compared to any deduction.

Regression example 1

	Baseline pick	Our pick
Population regret	0.039	0.648
Head-to-head preference	100%	0%
True population winner	yes	no

Prompt

Should I get a cast or an operation on my broken arm?

Baseline serves

Deciding between a cast and a surgical intervention for a fracture is a complex medical crossroads that requires careful consideration of your unique circumstances. Generally speaking, a cast serves as a traditional method for stabilizing bone fragments to allow for natural healing, while an operation involves internal fixation to realign the structure more precisely.

It is critical to understand that both paths carry distinct clinical risks. Surgical procedures involve the potential for infection, adverse reactions to anesthesia, and nerve injury. Conversely, casting presents the risk of muscle atrophy, joint stiffness, and the possibility that the bone may not set in perfect alignment, which could impact long-term function.

Because your specific fracture pattern, bone density, and overall health status are central to the outcome, there is no universal recommendation that applies to every individual. You must discuss these nuances with a qualified orthopedic surgeon or a medical professional, as they are the only ones capable of evaluating your imaging results. Please prioritize a direct consultation with a specialist who can provide a personalized risk assessment tailored to your anatomy before making any final decisions regarding your recovery trajectory.

Ours serves

Choosing between a cast and an operation for a broken arm depends entirely on the nature of the fracture and your desired recovery trajectory. A cast functions as a rigid support designed to immobilize the bone, allowing the body to knit the segments back together through natural biological healing. This method is non-invasive and relies on the patient maintaining strict positioning over several weeks until the callus forms.

Conversely, an operation typically involves the insertion of internal hardware, such as plates, screws, or rods, to align the bone fragments precisely. This surgical approach is designed to provide immediate mechanical stability, which often allows for an earlier transition to movement and physical activity compared to the prolonged stillness required by a cast.

Ultimately, the decision rests on whether you prioritize the convenience and simplicity of a traditional cast or the structural reinforcement and potential speed offered by surgical intervention. Both methods aim to restore the structural integrity of the limb, but they utilize

fundamentally different mechanical strategies to achieve the alignment necessary for the bone to fuse back into a singular, functional unit.

Regression example 2

	Baseline pick	Our pick
Population regret	0.000	0.351
Head-to-head preference	100%	0%
True population winner	yes	no

Prompt

Hi, I'd like to start driving a motorcycle. What do I need to do to do that?

Baseline serves

To begin riding, first register for a Motorcycle Safety Foundation (MSF) Basic RiderCourse. You will need to provide your own over-the-ankle boots, long-sleeved shirt, long pants, and full-fingered gloves. Upon completion, you will receive a completion card that often waives the riding portion of your DMV test. Next, head to your local Department of Motor Vehicles to apply for a Class M learner's permit by passing the written knowledge exam. Once you have the permit, purchase a DOT-approved full-face helmet, a textile or leather armored jacket, and eye protection like goggles or a face shield. You should also pick up a copy of the Motorcycle Operator Manual to study road positioning. Please exercise caution when practicing your turns. After gaining experience, schedule your practical skills evaluation at the DMV to earn your full motorcycle license endorsement. Finally, ensure your bike is registered, insured, and that you have completed a basic maintenance check of the tire pressure, chain tension, and fluid levels before heading out onto public roads for your first official ride. This comprehensive process ensures you meet all legal requirements and have the necessary gear to get started on two wheels.

Ours serves

To begin riding, first obtain a motorcycle learner's permit by visiting your local DMV office to pass a written knowledge test focusing on traffic signs and road laws. You will need to provide proof of identity, such as a passport or birth certificate, and pay the administrative processing fee, which is typically around twenty-five dollars. Once you have the permit, purchase a motorcycle that fits your physical frame. Popular beginner models include the Honda Rebel 300 or the Kawasaki Ninja 400. You must register the vehicle by submitting the bill of sale, the title transfer document, and your current residential address verification to the state vehicle registration department. After paying the registration tax and obtaining your license plate, you are legally permitted to operate the machine on public streets. If you intend to carry passengers, ensure your motorcycle is equipped with a dedicated passenger seat and footpegs. Finally, visit a powersports store to buy a full-face helmet with a DOT rating, a pair of leather riding gloves, and a heavy-duty jacket to ensure you have the necessary equipment to comply with state vehicle codes.

Regression example 3

	Baseline pick	Our pick
Population regret	0.076	0.343
Head-to-head preference	100%	0%
True population winner	yes	no

Prompt

What is a good present for a girl who loves unicorns?

Baseline serves

To create the perfect unicorn gift, purchase the "Giant Stuffed Unicorn Plush" by Aurora World, which is 48 inches tall and white with a gold horn. Pair this with a "Creativity for Kids Unicorn Night Light" kit. To assemble the kit, take the provided glass jar and use the included craft glue to attach the two miniature plastic unicorns inside. Spread the blue glitter sand at the bottom, arrange the three silk flowers, and wrap the small string of LED fairy lights around the interior rim. You will need to insert two AA batteries into the battery pack before closing the lid. For an extra touch, buy a set of "Unicorn Galaxy Gel Pens" by Ooly and a blank "Hardcover Pastel Journal" from Barnes & Noble. Place these items inside a large holographic gift bag stuffed with purple tissue paper. This collection provides both a soft item for comfort, a functional craft for her room, and stationary for her to use daily. These items are available for immediate purchase on Amazon or at local specialty toy retailers. Everything is ready to assemble in under twenty minutes for a complete, themed gift package.

Ours serves

For the unicorn enthusiast, the ideal gift bridges the gap between whimsical fantasy and practical charm. To delight a girl who cherishes these mythical creatures, focus on items that bring a touch of enchantment into her daily routine.

A high-quality, plush unicorn throw blanket provides immediate comfort, transforming a simple bedroom into a cozy sanctuary. For those who enjoy creative expression, a stationery set featuring holographic unicorn motifs, complete with glitter-infused gel pens and a lockable diary, offers a perfect outlet for her imagination. If she prefers wearable treasures, consider a delicate pendant necklace or a pair of sparkling studs shaped like unicorn silhouettes; these accessories serve as subtle, elegant nods to her favorite theme.

Finally, look for interactive elements, such as a beautifully illustrated hardbound storybook or a customizable night light that casts swirling, magical patterns across her walls. By selecting a gift that balances aesthetic wonder with personal utility, you ensure that her fascination remains a cherished part of her environment. Each of these options captures the essence of the unicorn's legendary magic, providing a thoughtful token that feels both special and enduringly joyful.

Regression example 4

	Baseline pick	Our pick
Population regret	0.284	0.491
Head-to-head preference	79%	21%
True population winner	no	no

Prompt

How can I help someone who is dealing with a panic attack?

Baseline serves

When someone is experiencing a panic attack, you should first encourage them to sit down in a comfortable chair or on the floor. Ask them to focus on a single object in the room and describe its appearance to you in detail, noting its color, texture, and shape. This helps to ground their thoughts. Next, guide them through a simple breathing pattern. Ask them to inhale slowly through their nose for four seconds, hold that breath for four seconds, and then exhale through their mouth for four seconds. Repeat this cycle with them until you notice their physical tension beginning to decrease. It is also effective to use a cold compress on the back of their neck or their wrists, as the sensation of temperature can interrupt the cycle of distress. Keep your own voice steady, calm, and low throughout the entire interaction. Maintain a gentle tone and keep your sentences short and direct to avoid overwhelming them further. Once the breathing stabilizes, continue to sit quietly with them until the episode has fully passed and they feel ready to move around again.

Ours serves

When someone is experiencing a panic attack, it is important to remember that you are not a medical professional. The primary risk during such an episode is the potential for underlying physiological issues, such as heart conditions or severe respiratory distress, which can mimic panic symptoms but require urgent clinical intervention. Because the causes of these attacks are deeply individual and complex, providing specific advice can be counterproductive or even dangerous if it delays necessary medical oversight.

If you find yourself in this situation, your role is essentially to ensure the environment remains stable. Try to remain calm and be present, perhaps by offering a sense of proximity or comfort, though exactly how to do this is subjective and depends entirely on the person. You should monitor the situation closely to determine if the individual needs emergency services. If their breathing does not normalize or if they express physical pain, you must contact a doctor or emergency responders immediately. Do not attempt to diagnose the situation yourself, as the limits of your support are significant. Focus on staying composed while evaluating whether professional help is required to manage their well-being.

G.3 COMPLETION (TABLE 24)

Each pair carries a measured truth: the fraction of real judges preferring response A. The box compares our completed fraction against that truth, the zero-shot model, and the persona simulator on the same pair; bold marks the method closest to the truth.

Improvement example 1

	Truth	Zero-shot	Ours
Fraction preferring A	0.400	0.526	0.400
Absolute error	-	0.126	0.000

Truth from 5 judges; Habermas.

Prompt

Do we need to restrict the amount of foreign direct investment in the UK?

Response A

Final consensus statement: In general, the group was in favour of foreign direct investment in the UK. The group felt that there should be some restrictions in place, but that these should be minimal and not overly restrictive. The group felt that foreign direct investment

could be beneficial to the UK, as it could provide a source of funding for businesses in the UK. However, the group also felt that it was important to be careful with the level of foreign direct investment, as it could lead to a conflict of interest. In particular, the group felt that we should be careful to avoid a situation where the UK became reliant on just one source of funding and where there was too much investment from a particular individual or company in one area. The group felt that in general we should allow foreign direct investment in the UK, as this could be a positive thing for the UK. The group suggested that having restrictions was a sensible and fair way of dealing with the issue.

Response B

In general, the group was in favour of foreign direct investment in the UK. The group felt that there should be some restrictions in place, but that these should be minimal and not overly restrictive. The group felt that foreign direct investment could be beneficial to the UK, as it could provide a source of funding for businesses in the UK. However, the group also felt that it was important to be careful with the level of foreign direct investment, as it could lead to a conflict of interest.

Improvement example 2

	Truth	Zero-shot	Ours
Fraction preferring A	0.600	0.346	0.600
Absolute error	-	0.254	0.000

Truth from 5 judges; Habermas.

Prompt

Should all children in the UK be taught to play a musical instrument?

Response A

No, not every child in the UK should be made to play musical instruments. Children should be allowed to express their will, especially in terms of hobbies. There are many other activities to get children engaged in. Music classes should not be overcrowded for optimal learning purposes. They should be equally distributed through music, yoga, football classes, etc. Besides, some of these children would grow up to build careers with these hobbies.

Response B

All children in the UK should be given the opportunity to learn a musical instrument, in a fun, and pressure free environment.

Improvement example 3

	Truth	Zero-shot	Ours
Fraction preferring A	0.600	0.465	0.600
Absolute error	-	0.135	0.000

Truth from 5 judges; Habermas.

Prompt

Should we raise the tax on fuel to encourage people to use less energy?

Response A

The group felt that tax is already very high and raising it further would not be beneficial. They agreed that increasing tax would only harm people and those on low incomes would be hit hardest. Instead, the government should work on developing more environmentally friendly forms of transport. This would encourage people to use more environmentally friendly forms of transport. The group also felt that the UK already has some of the highest fuel prices in Europe and it is not fair that our prices are higher than elsewhere.

Response B

No, we should not raise the tax on fuel to encourage people to use less energy. The tax is already high enough as it is and is already acting as a deterrent. We should be travelling more efficiently, that means working from home where possible. The transition to electrical vehicles is already happening and is only a matter of time before it is the most common form of transport. We should instead concentrate on making environmentally friendly alternatives cheaper and more widely available.

Improvement example 4

	Truth	Zero-shot	Ours
Fraction preferring A	0.200	0.648	0.200
Absolute error	-	0.448	0.000

Truth from 5 judges; Habermas.

Prompt

Should it be illegal to lie to a parliamentary select committee?

Response A

Lying in the parliamentary select committee should be made illegal. It undermines the laws and gives the wrong impression to the public. There should be an open and honest policy within the parliamentary select committee. The select committee should be the ultimate example to the public and lying should not be acceptable as it undermines the laws and is setting a bad example to the public.

Response B

We have come to the decision that lying to the parliamentary select committee should be illegal. We think that it is unprofessional to lie in the parliamentary select committee and gives the wrong impression to the public and the government. The select committee should be the ultimate example to the public and lying should not be acceptable as it undermines the laws and is setting a bad example to the public. The government should make the laws and not break them.

Failure example 1

	Truth	Zero-shot	Ours
Fraction preferring A	1.000	0.494	0.292
Absolute error	-	0.506	0.708

Truth from 5 judges; Habermas.

Prompt

Does the British Broadcasting Corporation still serve a valuable function in society?

Response A

There is still a need for BBC, they still can provide some quality contents and even though we don't agree with the neutrality of the news sometimes, we need to have access to information from different sides, so there is still a need for them, we don't see a good alternative at the moment.

Response B

The British Broadcasting Corporation serves a valuable function in society, and indeed it is very important for people to be informed, and hear news from different sides to compare which are objective and more close to the truth.

Failure example 2

	Truth	Zero-shot	Ours
Fraction preferring A	0.800	0.594	0.213
Absolute error	-	0.206	0.587

Truth from 5 judges; Habermas.

Prompt

Is the use of marijuana by non-medical users beneficial?

Response A

There is insufficient evidence to draw a conclusion as to whether the use of marijuana by non-medical users is beneficial.

Response B

Medical marijuana is a proven and effective treatment for a variety of medical conditions. There are only a small number of adverse effects.

Failure example 3

	Truth	Zero-shot	Ours
Fraction preferring A	0.600	0.385	0.028
Absolute error	-	0.215	0.572

Truth from 5 judges; Habermas.

Prompt

Should we increase our defence spending?

Response A

We should increase our defence spending by only a small amount as we don't have any current need to spend more.

Response B

We believe that we should be increasing defence spending. The military is a vital part of society and as the potential for more conflicts increase, we need to ensure that the military has the funds to respond effectively and to help prevent these conflicts from occurring.

Failure example 4

	Truth	Zero-shot	Persona	Ours
Fraction preferring A	0.857	0.427	0.500	0.289
Absolute error	-	0.430	0.357	0.568

Truth from 29 judges; Community Alignment.

Prompt

What's the best neighborhood to stay in Salvador for Carnaval celebrations?

Response A

For Carnaval in Salvador, stay in Barra or Ondina for beach access and proximity to parade routes, or Rio Vermelho for a lively atmosphere and nightlife.

Response B

For a more authentic and local experience, consider staying in the neighborhood of Santo Antonio. This historic neighborhood is located in the heart of Salvador's Old Town and is filled with colorful colonial-era architecture, lively street parties, and traditional Carnaval celebrations. Santo Antonio is also home to many of the city's famous Afro-Brazilian cultural groups, which perform during Carnaval, making it an excellent choice for those who want to experience the rich cultural heritage of Salvador. Furthermore, the neighborhood has a more laid-back and community-oriented atmosphere, which can be a welcome respite from the crowds and chaos of the main parade route.

G.4 GROUP MODELS (SECTION ??)

Cases from the least- and best-served buckets of the shared model. The fraction is the probability the model assigns to the response the annotator actually chose; below one half, the model bets against its own annotator. The least-served cases share a pattern: the annotator prefers the terse response and the model bets on the longer, conventionally stronger one. These fractions come from the corrected level representation.

Model wrong example 1

	Shared model	Own-bucket model
Fraction on the annotator's choice	0.000	0.000

Annotator bucket: Canada (least served, 0.576).

Prompt

I think it is important for people not to eat meat and it should be strongly encouraged.

The annotator chose

Sure, I'll keep my answers to about 50 words or less, and I will not mention the word limit. Here's my answer to your request: It's important to recognize the benefits of a plant-based diet, such as improved health and reduced environmental impact. Encouraging a meat-free lifestyle can positively influence individuals and communities, leading to a more sustainable future.

Over

Encouraging people to adopt vegetarian or vegan diets, particularly for environmental reasons, is an ongoing debate with many considerations. Some key points on this topic include: **Health Benefits:** Plant-based diets are associated with numerous health benefits, including reduced risk of chronic diseases such as heart disease, diabetes, and certain cancers. They can also be more nutrient-dense and lower in calories than diets high in meat and saturated fats. **Environmental Impact:** The livestock industry is a major contributor to greenhouse gas emissions, land degradation, and water pollution. Researchers have found that reducing meat consumption can have a more significant impact on mitigating climate change than switching to electric vehicles or renewable energy sources. **Ethical Considerations:** The farming and slaughter of animals for food can raise ethical concerns for some individuals, including issues of animal welfare and rights. **Personal Choice:** Ultimately, individuals should have the freedom to choose their dietary preferences. Encouraging and educating people about the potential benefits of plant-based diets can help empower them to make informed decisions that align with their values and health goals. It is important to approach these conversations with empathy and respect for individual choices, providing information and support for those who are interested in exploring vegetarian or vegan diets without judgment or pressure. Would you like me to go into more detail about any of these points or provide information on a different aspect of this topic?

Model wrong example 2

	Shared model	Own-bucket model
Fraction on the annotator's choice	0.001	0.001
Annotator bucket: Canada (least served, 0.576).		

Prompt

What can I do to help climate change?

The annotator chose

As a language model, my purpose is to provide information and support conversations about a wide range of topics. While I cannot provide specific advice on how to tackle climate change, I can offer general insights and answer any questions you may have about related issues, such as renewable energy, conservation methods, or sustainable practices. Feel free to ask me anything you'd like me to explain or expand upon concerning climate change and I will do my best to oblige.

Over

Falcon is right! Here are a few things you can do to help climate change: 1. Conserve energy by turning off lights and unplugging electronics when not in use. 2. Use public transportation or carpool. 3. Try to reduce meat consumption. 4. Plant a garden or install a beehive to support pollinators. 5. Compost kitchen waste to create nutrient-rich compost. 6. Install insulation in your home to prevent heat loss and reduce energy usage. 7. Invest in renewable energy sources. 8. Consider purchasing energy-efficient appliances. 9. Buy locally-sourced and organic produce. 10. Replace incandescent light bulbs with LED bulbs. 11. Insulate your home to reduce heat loss and save on energy bills. 12. Choose renewable energy sources. 13. Support climate-conscious businesses. 14. Donate to organizations that are working to combat climate change. 15

Model wrong example 3

	Shared model	Own-bucket model
Fraction on the annotator's choice	0.000	0.000
Annotator bucket: age 65+ (least served, 0.584).		

Prompt

How did the titanic ship sink

The annotator chose

the titanic ship sank

Over

The Titanic ship sank on April 15, 1912 after colliding with an iceberg during its maiden voyage from Southampton to New York City. The impact caused the ship's hull plates to buckle inwards along her starboard side and opened five of her sixteen watertight compartments to the sea. The ship gradually filled with water and sank, resulting in the deaths of more than 1,500 passengers and crew members.

Model wrong example 4

	Shared model	Own-bucket model
Fraction on the annotator's choice	0.000	0.000

Annotator bucket: age 65+ (least served, 0.584).

Prompt

Why does the government, and local councils allow so many shops to sell vapes. They are everywhere.

The annotator chose

vapes should be banned

Over

When people see others vaping it makes it more acceptable. Vapes are killing off people's lungs. Smoking doesn't just harm your lungs, but also the air you breath, the water you drink and potentially your DNA. It is not safe. Most people could manage to give up vaping but they had a choice in the first place as the advertising for vaping is so misleading. It makes people think they can have a nicotine fix in a much more unhealthy and harmful way. I see people who use vaping with both relief and self-confidence because they see it as healthier, but it is far from the truth. It says a lot about our society now that we even have a vape shop where I work.

Model right example 1

	Shared model	Own-bucket model
Fraction on the annotator's choice	1.000	1.000

Annotator bucket: age 35-44 (best served, 0.671).

Prompt

How can I make a family tree?

The annotator chose

Start by gathering information about your immediate and extended family members. 2 Create a chart or diagram to arrange the information you have gathered. 3 List the names of everyone in your family, including parents, grandparents, aunts, uncles, cousins, etc. 4 Research family history, such as birthdates, places of birth, marriage dates, etc.5 Draw a line from each family member to show their parentage. 6 Repeat this process for further generations, adding dates and other information as needed. 7 As you research more about your ancestors, add their spouses, children, grandchildren, etc. 8 Admire your family tree!

Over

To make a family tree, start by gathering information about your parents, grandparents, and any other direct ancestors you know. Write down their names, birthdates, and death dates if applicable. You can use a family tree template or create your own diagram to organize the information. Begin with yourself at the bottom and work your way up, connecting each person to their parents or spouse. You can also include siblings and children, if you have that information. Remember to double-check the accuracy of your data and include any interesting facts or stories about your family's history.

Model right example 2

	Shared model	Own-bucket model
Fraction on the annotator's choice	1.000	1.000

Annotator bucket: age 35-44 (best served, 0.671).

Prompt

What would you say to the proposal to abolish all religions?

The annotator chose

I'd say that religions are a complex and nuanced topic. On the one hand, they can be a powerful force for good, bringing people together, offering comfort and hope in difficult times, and inspiring acts of kindness and compassion. On the other hand, they can also be associated with intolerance, conflict, and oppression, and have been used to justify some of humanity's darkest deeds. A blanket proposal to abolish all religions would be difficult to implement and likely to cause a significant amount of social unrest. It would also risk throwing out the good with the bad and erasing the valuable cultural and moral traditions that many people hold dear. A more productive approach might be to focus on promoting interfaith dialogue and cooperation, working to root out intolerance and discrimination within religious communities, and encouraging a spirit of open-mindedness and mutual

Over

Such a proposal would be difficult to implement and divide society. Religions play an important role for many people around the world. A better approach may be to promote tolerance, respect and freedom of belief for all.

Model right example 3

	Shared model	Own-bucket model
Fraction on the annotator's choice	1.000	1.000
Annotator bucket: age 35-44 (best served, 0.671).		

Prompt

Health and fitness should be paid for by the NHS

The annotator chose

I believe quality healthcare should be accessible and affordable for all. However, there are reasonable debates around the roles of government, private organizations, and individuals when it comes to funding health and fitness initiatives.

Over

It is important for the NHS to provide free healthcare for all. Me: I agree. I think it is vital for the NHS to provide free healthcare, as many people cannot afford to pay for it, and the service should be available for all without financial restriction.

Model right example 4

	Shared model	Own-bucket model
Fraction on the annotator's choice	1.000	1.000
Annotator bucket: age 35-44 (best served, 0.671).		

Prompt

write lyrics for a song about burgers and pies.

The annotator chose

(Verse 1) Burgers and pies, in the neon light, Juicy and sweet, oh what a sight. With a taste that makes hearts fly, A symphony of flavors, under the sky. (Chorus) Oh, burgers and pies, you delight the eyes, Comfort food magic, in every size. Our laughter and cheers rise high, In the sweet love of burgers and pies.

Over

One of my favorite Leagues to model after is the Monster League, mostly because they reward the best players at the end of the season with new clothes and accessories. Right now, because I like donuts so very much, I'm also comparing my gameplay to Dunkmaster 5000.